



Molecular Genetics

CHAPTER 3

EXTENSIONS TO MENDEL'S LAW

References:

Doctor Osama Records,
and The book : From Gene to genome 5th edition

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Chapter 3: Extensions to Mendel's laws

Some phenotypic variation poses a challenge to Mendelian analysis

UNLIKE the pea traits that Mendel examined, most human characteristics do not fall neatly into just two opposing phenotypic categories.

→ These complex traits, such as skin and hair color, height, athletic ability, and many others (The same can be said of traits expressed by many of the world's food crops: Their size, shape, succulence, and nutrient content vary over a wide range of values) seem to defy Mendelian analysis.

Lentils for example come in an intriguing array of colors and patterns (Fig. 3.1). crosses between pure-breeding lines of lentils result in some startling surprises. A cross between pure-breeding tan and pure-breeding gray parents, for example, yields an all-brown F1 generation. When these hybrids self-pollinate, the F2 plants produce not only tan, gray, and brown lentils, but also green.

على عكس التشابتر الماضي الي بحكي عن قوانين مندل.. هاد التشابتر بحكي عن صفات ما بتخضع لقوانين مندل وما يكون الها بس 2 phenotypes عكس بعض لهيك رح نتوسع اكثر بهاد التشابتر لنحكي عن وراثه الصفات المعقدة مثل لون الشعر، الطول، شكل وحجم بعض المحاصيل الزراعيه وكثير صفات تانية الها احتمالات اوسع من وراثه مندل.

مثال بسيط على الموضوع هو لون حبوب العدس. لما نعمل crossing بين pure-

breeding tan and pure-breeding gray parents offspring ما بطلعوا tan و gray او كلو tan او gray بل بطلع لونهم brown. ويرضو اذا عملنا self-pollination لل brown بطلع ال F2: gray, tan, brown and green!!!

So according to what is said previously, Crosses of pure-breeding lines can result in progeny phenotypes that don't appear to follow Mendel's rules

- No definitively dominant or recessive allele
- More than two alleles exist
- Multiple genes involved
- Gene-environment interactions



In this array of green, brown, and red lentils, some of the seeds have speckled patterns, while others are clear.
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Figure 3.1 Some phenotypic variation poses a challenge to Mendelian analysis. Lentils show complex speckling patterns that are controlled by a gene that has more than two alleles

Concept: 3.1 Extensions to Mendel for *single-gene* inheritance

A. Dominance is NOT always complete

- When defining dominance and recessiveness, we depend on the **F1** hybrids that arise from a mating between two pure-breeding lines
 → If a hybrid is identical to one parent for the trait under consideration, the allele carried by that parent is deemed dominant to the allele carried by the parent whose trait is not expressed in the hybrid.

For example: if a mating between a pure-breeding white line and a pure-breeding blue line produces F1 hybrids that are white, the white allele of the gene for color is dominant to the blue allele. If the F1 hybrids are blue, the blue allele is dominant to the white one

Figure 3.2 summarizes different kinds of dominance Mendel had described. It diagrams two situations

in which neither allele of a gene is completely dominant.

As the figure shows, crosses between true-breeding strains can produce hybrids with **phenotypes that differ from both parents.**

Figure 3.2 Different dominance relationships. The phenotype of the heterozygote defines the dominance relationship between two alleles of the same gene (here, A^1 and A^2). Dominance is complete when the hybrid resembles one of the two pure-breeding parents. Dominance is incomplete when the hybrid resembles neither parent; its novel phenotype is usually intermediate. Codominance occurs when the hybrid shows traits from both pure-breeding parents.

Type of Dominance	A^1A^1	A^2A^2	A^1A^2 hybrids	
Complete				A^1 is dominant to A^2 A^2 is recessive to A^1
Complete				A^2 is dominant to A^1 A^1 is recessive to A^2
Incomplete				A^1 and A^2 are incompletely dominant relative to each other
Codominant				A^1 and A^2 are codominant relative to each other

لما نكون بدنا نعرّف السيادة (dominancy) والتنحي (recessiveness) منعتمد بشكل أساسي على hybrids الجيل الأول (F1) هون تعني الفرد الناتج من تزاوج parents عندهم شكلين مختلفين لنفس الصفة). اذا ال hybrid كان مطابق لأحد الابوين بالصفة الي عم يتم دراستها، ف ال allele الي حامله هاد ال parent هو معتبره dominant على ال allele الثاني الي حامله ال parent صاحب الصفة الي ما ظهرت.

هلا خلينا نبسط Figure 3.2 الي ملخصنا موضوع ال dominancy.

اول حالة: تزاوج الابيض (A^1A^1) مع الازرق (A^2A^2) أنتجنا افراد ال F1 كلهم أبيض ← الاليل A^1 هو ال dominant
ثاني حالة: تزاوج الابيض (A^1A^1) مع الازرق (A^2A^2) أنتجنا افراد ال F1 كلهم أزرق ← الاليل A^2 هو ال dominant
 ← أول حالتين يمثلوا ال complete dominance

ثالث حالة: تزاوج الابيض (A^1A^1) مع الازرق (A^2A^2) أنتجنا افراد ال F1 بصفة متوسطة بين اللونين وما يمثل أي من ال parents ← A^1 and A^2 are **incompletely** dominant to each other يعني ولا واحد فيها dominant على الثاني
رابع حالة: تزاوج الابيض (A^1A^1) مع الازرق (A^2A^2) أنتجنا افراد ال F1 بحملوا الصفتين ال parents مع بعض ← A^1 and A^2 are **codominant** relative to each other

➔ In simple words:

Complete dominance: Hybrid resembles one of the two parents

Incomplete dominance: Hybrid resembles neither parent

Codominance: Hybrid shows traits from both parents

وبعد ما خدنا فكرة عن هذول الحالتين (الثالثة والرابعة) خلينا نبش نشرحهم بالتفصيل

1- Incomplete dominance: The F₁ hybrid resembles neither pure-breeding parent

the hybrid **does not** resemble either pure-breeding parent. F₁ hybrids that differ from both parents often express a phenotype that is intermediate between those of the pure-breeding parents.

→ with incomplete dominance, neither parental allele is dominant nor recessive to the other; both contribute to the F₁ phenotype

▪ Examples:

- Blooming time: A cross between pure late-blooming and pure early-blooming pea plants results in an F₁ generation that blooms in between the two extremes.
- Flower color in many plant species like **snapdragons**. a cross between pure-breeding red-flowered parents and pure-breeding white yields hybrids with pink blossoms, as if a painter had mixed red and white pigments to get pink (Fig. 3.3a).
If allowed to self-pollinate, the F₁ pink-blooming plants produce F₂ progeny bearing red, pink, and white flowers in a ratio of **1:2:1** (Fig. 3.3b).

▪ Biochemical explanation:

each allele of the gene under analysis specifies an alternative form of a protein molecule with an enzymatic role in red pigment production. The white allele (A^2) does not give rise to a functional enzyme, but the red allele (A^1) does.

Thus:

→ two red alleles per cell ($A^1 A^1$) produce a double dose of a red producing enzyme, which generates enough pigment to make the flowers look fully red

→ In the heterozygote ($A^1 A^2$), one copy of the red allele per cell results in only enough pigment to make the flowers look pink

→ In the homozygote for the white allele ($A^2 A^2$), where there is no functional enzyme and thus no red pigment, the flowers appear white.

A^1 gene هو الي بعطي الانزيم المُنتج للون الاحمر... ف لما يكونوا الـ two alleles الـ A^1 فهاد يعني انو رح يكون عندي جرعتين من اللون الاحمر فبعطيني لون أحمر. أما لما يكون بس $(A^1 A^2)$ فهاد يعني انو عندي جرعة وحدة من اللون الاحمر لهيك بطلع لونو فاتح فيتظهر الوردة باللون الزهري. واخر حالة هي لما ما يكون في A^1 بالمرّة (يعني $A^2 A^2$) ف رح يطلع عندي اللون أبيض لأنو ما في انتاج للون الاحمر ابدا

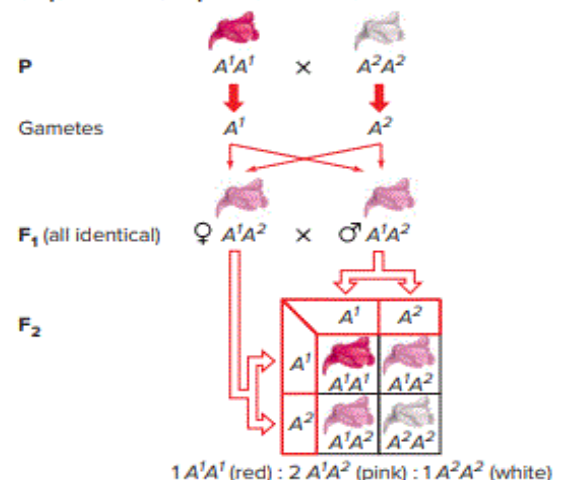
Figure 3.3 Pink flowers are the result of incomplete dominance. (a) Color differences in these snapdragons reflect the activity of one pair of alleles. (b) The F₁ hybrids from a cross of pure-breeding red and white strains of snapdragons have pink blossoms. Flower colors in the F₂ appear in the ratio of 1 red : 2 pink : 1 white. This ratio signifies that the alleles of a single gene determine these three colors.

a: © Henry Hemming/Getty Images RF

(a) *Antirrhinum majus* (snapdragons)



(b) A Punnett square for incomplete dominance



2- Codominance: The F₁ hybrid exhibits traits of both parents

the progeny looks like **both** parents

→ Both traits show up equally in the heterozygote's phenotype which means that neither allele is dominant or recessive to the other

Examples:

- Lentils (Fig. 3.4a): A cross between pure-breeding spotted (بقعة) lentils and pure breeding dotted (نقطة) lentils produces heterozygotes that are both spotted and dotted.

Self-pollination of the spotted/dotted F₁ generation generates F₂ progeny in the ratio of 1 spotted : 2 spotted/dotted : 1 dotted

- Membrane-anchored molecules that distinguish different types of red blood cells (Fig. 3.4b): one gene (*I*) with alleles *I*^A and *I*^B controls the presence of a sugar polymer that protrudes from the red blood cell membrane.

→ *I*^A *I*^A: sugar type A → A blood type

→ *I*^A *I*^B: Sugar type A & B → AB blood type

→ *I*^B *I*^B: Sugar type B → B blood type

Biochemical explanation:

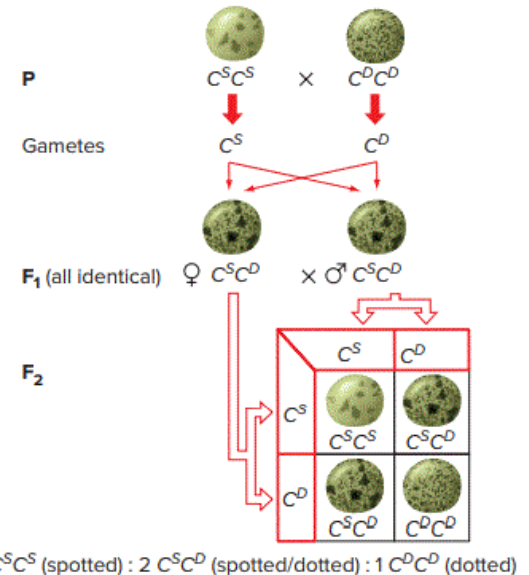
Each of the alternative alleles encodes a slightly different form of an enzyme that causes production of a slightly different form of the complex sugar

→ So, when both alleles produce a functional gene product, they are usually codominant for phenotypes analyzed at the molecular level.

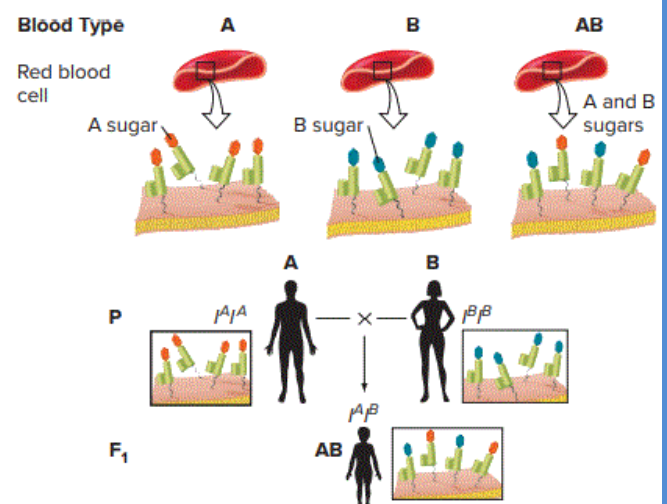
- ❖ In incomplete dominance and codominance, the ratio 1:2:1 is the familiar genotypic ratio of an ordinary single-gene F₁ self-cross. What is new is that because the heterozygotes **look unlike either homozygote**, the phenotypic ratios are an exact reflection of the genotypic ratios.

Figure 3.4 In codominance, F₁ hybrids display the traits of both parents. (a) A cross between pure-breeding spotted lentils and pure-breeding dotted lentils produces heterozygotes that are both spotted and dotted. Each genotype has its own corresponding phenotype, so the F₂ ratio is 1:2:1. (b) The *I*^A and *I*^B blood group alleles are codominant because the red blood cells of an *I*^A*I*^B heterozygote have both kinds of sugars at their surface.

(a) Codominant lentil coat patterns



(b) Codominant blood group alleles



لما كان مندل يزواج الازهار البنفسجية مع البيضاء (حالة الcomplete dominance) .. كان يطلع معو ال F_1 كلو بنفسجي وبعدها لما يعمل self-pollination لل F_1 كان يطلع معو 3 بنفسجي ووحدة بيضاء .. هاي النسبة (3:1) منسميها ال *phenotypic ratio* بس ال *genotypic ratio* كانت مختلفة .. هاي النسبة كانت 1:2:1 (PP,Pp,Pp,pp) فالاختلاف بين الي كان يطلع مع مندل بال complete dominance) وال incomplete and codominance هو انو ال *phenotypic ratios* وال *genotypic ratios* فيهم هي نفسها ← 1:2:1 (الheterozygote طللعو phenotype جديد ما يشبه الparents)

- Dominance relations between alleles do NOT affect transmission of alleles

Type of dominance (complete, incomplete dominance, codominance) depends on:

1- the type of **proteins encoded**

2- the **biochemical functions** of the proteins

→ so, variation in dominance relations do not negate Mendel's laws of segregation. Rather, they reflect differences in the way gene products control the production of phenotypes adding a level of complexity to the tasks of interpreting the visible results of gene transmission and inferring genotype from phenotype.

B. A Gene May Have More Than Two Alleles

Mendel analyzed either-or traits controlled by genes with two alternative alleles, but for many traits, more than two alternatives exist.

- Multiple alleles of a gene can segregate in populations
- Each individual can carry only two alleles
- Dominance relations are always **relative** to a second allele and are unique to a pair of alleles

(دائما لما نحكي عن الdominancy منحكي عن البولين مع بعض مو واحد .. ف منحكي الاليل A هو dominant على B مو "الاليل A هو dominant" وبس .. لازم نحدد هو dominant على مين ف دائما تزكروا انو تذكروا الpair الي عم نحكي عنهم بالزبط بالDominance relations)

Here, we look at three such traits: **human ABO blood types, lentil seed coat patterns, and human histocompatibility antigens.**

1- ABO blood types If a person with blood

The gene for the ABO blood types has three alleles: I^A , I^B , and i (Fig. 3.5a).

I^A → gives rise to blood type A by specifying an enzyme that adds sugar A

I^B → I^B give rise to blood type B by specifying an enzyme that adds sugar B

i → does not produce a functional sugar-adding enzyme

- Dominance relations are relative to a second allele:
 - I^A and I^B are codominant
 - I^A and I^B are dominant to i
- ➔ Blood type O is therefore a result of homozygosity for allele i .
- The phenotype A is a result of : $I^A I^A$ OR $I^A i$
- The phenotype B is a result of: $I^B I^B$ OR $I^B i$
- The phenotype O is a result of : ii ONLY

Figure 3.5 ABO blood types are determined by three alleles of one gene. (a) Six genotypes produce the four blood group phenotypes. (b) Blood serum contains antibodies against foreign red blood cell molecules. (c) If a recipient's serum has antibodies against the sugars on a donor's red blood cells, the blood types of recipient and donor are incompatible, and coagulation of red blood cells will occur during transfusions. In this table, a plus (+) indicates compatibility, and a minus (−) indicates incompatibility. Antibodies in the donor's blood usually do not cause problems because the amount of transfused antibody is small.

(a)	Genotypes	Corresponding Phenotypes: Type(s) of Molecule on Cell
	$I^A I^A$ $I^A i$	A
	$I^B I^B$ $I^B i$	B
	$I^A I^B$	AB
	ii	O

(b)	Blood Type	Antibodies in Serum
	A	Antibodies against B
	B	Antibodies against A
	AB	No antibodies against A or B
	O	Antibodies against A and B

(c)	Blood Type of Recipient	Donor Blood Type (Red Cells)			
		A	B	AB	O
	A	+	−	−	+
	B	−	+	−	+
	AB	+	+	+	+
	O	−	−	−	+

- ✦ several conclusions from these observations can be drawn:
 - First: a given gene may have more than two alleles, or multiple alleles; in our example, the series of alleles is denoted I^A , I^B , and i .
 - Second: although the ABO blood group gene has three alleles, each person carries only two of the alternatives— $I^A I^A$, $I^B I^B$, $I^A I^B$, $I^A i$, $I^B i$, or ii . Thus, six possible ABO genotypes exist. (Because each individual carries no more than two alleles for each gene, no matter how many alleles are in a series, Mendel's law of segregation remains intact, because in a sexually reproducing organism, the two alleles of a gene separate during gamete formation.)
 - Third, an allele is not inherently dominant or recessive; its dominance or recessiveness is always relative to a second allele.

Medical and legal implications of ABO blood group genetics:

- Medical: Matching ABO blood types is a prerequisite of successful blood transfusions, because people make antibodies to foreign blood cell molecules:
 - A person whose cells carry only A molecules → produces anti-B antibodies
 - B people → manufacture anti-A antibodies
 - AB individuals → make neither type of antibody → **universal recipients**
 - O individuals → produce both anti-A and anti-B antibodies → **universal donor** (Fig. 3.5b).
(عشان هيك ما بياخد دم الا من الي زيو)
 (These antibodies cause coagulation of cells displaying the foreign molecules (Fig. 3.5c))
- Legal: Information about ABO blood types can also be used as legal evidence in court, to exclude the possibility of paternity or criminal guilt like what happened in 1944.
The actress Joan Barry (phenotype A) sued Charlie Chaplin (phenotype O) for support of a child (phenotype B) whom she claimed he fathered. The scientific evidence indicated that Chaplin could not have been the father, since he was apparently ii and did not carry an I^B allele. This evidence was admissible in court, but the jury was not convinced, and Chaplin had to pay.

2- Lentil seed coat patterns

Five alleles for C (coat color) gene: spotted (C^S), dotted (C^D), clear (C^C), marbled-1 (C^{M1}), and marbled-2 (C^{M2})

هاد المثال بوضحنا كثير منيچ المقصود بانو ال dominance وال recessiveness بتتحكى ل pair من ال alleles مو لواحد بس

- Reciprocal crosses between pairs of pure-breeding lines of all patterns (marbled-1 × marbled-2, marbled-1 × spotted, marbled-2 × spotted, and so forth) have clarified the dominance relations of all possible pairs of the alleles to reveal a **dominance series in which alleles are listed in order from most dominant to most recessive**.

عملنا تزاوج بين كل أليلين وطلع معنا شو ال dominance relations الي بتجمعهم وعرفنا ما يعرف بال dominance series وهي طريقة برتب فيها ال alleles من الأعلى dominant لأكثر واحد recessive

→ for ex. crosses of marbled-1 with marbled-2, or of marbled-1 with spotted or dotted or clear, produce the marbled-1 phenotype in the F_1 generation and a ratio of three marbled-1 to one of any of the other phenotypes in the F_2 . These results indicate that the marbled-1 allele is completely dominant to each of the other four alleles.

ف زي ما هو موضح بالشكل 3.6 لما نعمل تزاوج بين ال marbled-1 أو marbled-2 مع أي واحد من هذول : Spotted, dotted, or clear, ديامل ال F_1 بكونوا Marbled-1 أو marbled-2 (حسب الي زواجته) و هاد يعني انو ال marbled-1 & 2 همة dominant على كل الباقيين

ولكن..لما نعمل تزاوج بين ال marbled-1 و marbled-2 بطلعوا ال F₁ marbled-1 وهاد يعني انو marbled-1 هو dominant على ال marbled-2 والتزاوج بين ال dotted او ال spotted مع ال clear بطلع ال F₁ كلهم dotted او spotted وهاد يعني انهم dominant على ال clear ف بتطلع عندي ال dominance series بالشكل هاد:



So dominance relations are meaningful only when comparing two alleles: An allele, such as marbled-2, can be recessive to a second allele (marbled-1) but dominant to a third and fourth (dotted and clear). The fact that all tested pairings of lentil seed coat pattern alleles yielded a 3:1 ratio in the F₂ generation (except for spotted × dotted, which yielded the 1:2:1 phenotypic ratio reflective of **codominance**) indicates that these lentil seed coat patterns are determined by different alleles of the same gene.

Figure 3.6 How to establish the dominance relations between multiple alleles. Pure-breeding lentils with different seed coat patterns are crossed in pairs, and the F₁ progeny are self-fertilized to produce an F₂ generation. The 3:1 or 1:2:1 F₂ monohybrid ratios from all of these crosses indicate that different alleles of a single gene determine all the traits. The phenotypes of the F₁ hybrids establish the dominance relationships (bottom). Spotted and dotted alleles are codominant, but each is recessive to the marbled alleles and is dominant to clear.

3- Human Histocompatibility antigens

In some multiple allelic series, each allele is **codominant** with every other allele, and every distinct genotype therefore produces a distinct phenotype → This happens particularly with traits defined at the molecular level

Example: group of three major genes (HLA-A, HLA-B, and HLA-C) encode a family of related cell surface molecules in humans and other mammals known as **histocompatibility antigens** → location: Cell surface molecules present on all cells except RBCs and sperm

→ Role: Facilitates proper immune response to foreign antigens (e.g. virus or bacteria) while leaving the body's own tissues intact each of the three major histocompatibility genes has between 400 and 1200 alleles

- Each allele is codominant to every other allele
- Every genotype produces a distinct phenotype
- Enormous phenotypic variation

Parental Generation Parental seed coat pattern in cross Parent 1 × Parent 2	F ₁ Generation F ₁ phenotype	F ₂ Generation Total F ₂ frequencies and phenotypes	Apparent phenotypic ratio
marbled-1 × clear	marbled-1	798 296	3 : 1
marbled-2 × clear	marbled-2	123 46	3 : 1
spotted × clear	spotted	283 107	3 : 1
dotted × clear	dotted	1706 522	3 : 1
marbled-1 × marbled-2	marbled-1	272 72	3 : 1
marbled-1 × spotted	marbled-1	499 147	3 : 1
marbled-2 × dotted	marbled-2	182 70	3 : 1
spotted × dotted	spotted/dotted	168 339 157	1 : 2 : 1

Dominance series: marbled-1 > marbled-2 > spotted = dotted > clear

- ➔ no two people are likely to carry the same array of histocompatibility antigens on the surfaces of their cells (except for monozygotic twins)

C. Mutations Are the Source of New Alleles

Mutations: chance alterations of the genetic material that arise spontaneously in nature.

➔ If they occur in **gamete-producing cells**, they can be transmitted to offspring

- The frequency of gametes carrying a new mutation in a particular gene varies anywhere from 1 in 10,000 to 1 in 1,000,000 (This range exists because different genes have different mutation rates)
- Mutations that result in phenotypic variants can be used by geneticists to follow gene transmission (like, if all peas were yellow, Mendel would not have been able to decipher the transmission patterns of the gene for the seed color trait)

• Allele frequencies and monomorphic genes

Each organism carries two copies of every gene. So, to calculate the number of copies of a gene in a given population we multiply the number of individuals by 2.

➔ Each allele of the gene accounts for a percentage of the total number of gene copies, and that percentage is known as the **Allele Frequency**.

- Wild-type alleles: The most common alleles in a population, often designated by a superscript plus sign (+).

An allele is considered wild-type if it is present in the population at a frequency greater than 1%

- Mutant allele: a rare allele in the population

➔ the definitions of wild-type versus mutant alleles **are not static**. A newly induced mutation generates a mutant allele whose frequency can increase and over time, the allele can become wild-type.

- **Monomorphic:** Gene with only one common wild-type allele

Example: Agouti coat color gene in mice.

The wild-type allele (A) produces fur with each hair having yellow and black bands that blend together from a distance to give the appearance of dark gray, or *agouti*

Researchers have identified in the laboratory 14 distinguishable mutant alleles for the agouti gene.

➔ (a^t) : recessive to the wild type and gives rise to a black coat on the back and a yellow coat on the belly

➔ (a): also recessive to A and produces a pure black coat (Fig. 3.7).

- a recessive to A
- a^t dominant to a but recessive to A
 - In nature, wild-type agoutis (AA) survive to reproduce
 - But very few black-backed or pure black mutants ($a^t a^t$ or aa) do so because their dark coat makes it hard for them to evade the eyes of predators.

→ As a result, A is present at a frequency of much more than 99% and is thus the only wild-type allele in mice for the *agouti* gene

- **Polymorphic:** Gene with more than one common allele
 → High-frequency alleles of polymorphic genes are called **Common variants**
 Example:
 1. ABO blood type system: all three alleles— I^A , I^B , and i —have appreciable frequencies in most human populations.
 2. HLA histocompatibility genes

D. One Gene May Contribute to Several Characteristics

Pleiotropy: The phenomenon of a single gene determining a number of distinct and seemingly unrelated characteristics

How pleiotropy arise? Each gene determines a specific protein (or RNA) and because of that, each gene product can have a cascade of effects on an organism

Example:

- 1- Maori people of New Zealand: Many men exhibit a **syndrome** (a group of problems that are usually seen together) in which they develop respiratory issues but are also sterile
 Explanation: Researchers have found that the fault lies with the recessive allele of a single gene.

The gene's normal dominant allele specifies a protein necessary for the action of cilia and flagella → both of which are hairlike structures extending from the surfaces of some cells.

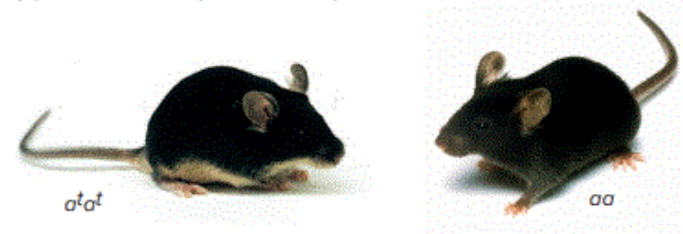
In men who are homozygous for the recessive allele, cilia that normally clear the airways fail to work effectively, and flagella that normally propel sperm fail to do their job (immotile). Thus, one gene determines a protein that affects both respiratory function and reproduction.

- ❖ Because most proteins act in a variety of tissues and influence multiple biochemical processes, mutations in almost any gene may have pleiotropic effects.

Figure 3.7 The mouse *agouti* gene: One wild-type allele, many mutant alleles. (a) Black-backed, yellow-bellied (top left); black (top right); and agouti (bottom) mice. (b) Genotypes and corresponding phenotypes for alleles of the *agouti* gene. (c) Crosses between pure-breeding lines reveal a dominance series. Interbreeding of the F_1 hybrids (not shown) yields 3:1 phenotypic ratios of F_2 progeny, indicating that A, a^t , and a are in fact alleles of one gene.

a (top left): © McGraw-Hill Education. Jill Birschbach, photographer. Arranged by Alexandra Dove, McArdle Laboratory, University of Wisconsin-Madison; a (top right, bottom): © Charles River Laboratories

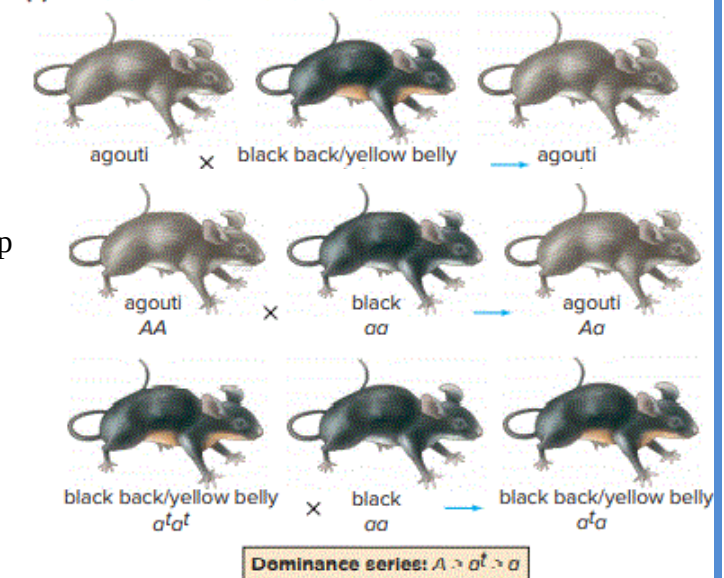
(a) *Mus musculus* (house mouse) coat colors



(b) Alleles of the *agouti* gene

Genotype	Phenotype
A-	agouti
$a^t a^t$	black/yellow
aa	black
$a^t a$	black/yellow

(c) Evidence for a dominance series



→ With some pleiotropic genes

- Heterozygotes can have a visible phenotype
- Homozygotes can be **lethal**

Example: A^Y allele of agouti gene in mice (Fig 3.8):

One of the 14 known mutant alleles of the agouti gene, A^Y → gives rise to mice with a much lighter, almost yellow color.

- When pure-breeding AA mice are mated to yellow mice, the offspring always emerge in a **1:1 ratio** of the two coat colors (Fig. 3.8a).

→ From this result, we can draw three conclusions:

- (1) All yellow mice must carry the wildtype A allele even though they do not express the agouti phenotype
- (2) yellow is therefore dominant to agouti
- (3) all yellow mice are $A^Y A$ heterozygotes.

(لأنهم $A^Y A^Y$ وال $A^Y A$ على الـ A كان ما رح يطلع
(ولا واحد agouti)

→ A^Y is dominant to A for hair color,
but is recessive to A for lethality

- Cross Yellow x Yellow mice:

- F_1 mice are 2/3 yellow and 1/3 agouti

→ This 2:1 ratio is indicative of a recessive lethal allele

- Pure-breeding yellow ($A^Y A^Y$) mice cannot be obtained because they are NOT viable

- ❖ An allele, such as A^Y , that negatively affects the survival of a homozygote is known as a **recessive lethal allele** (لأنه محتاج نسختين من الأليل عشان يكون مميت)

Delayed Lethality:

بالمثال الي ناقشناه سابقا.. كان الـ mouse الحامل للـ $A^Y A^Y$ alleles يموت قبل الولادة ← **Prenatally** (in utero)
ولكن ببعض الـ mutations ممكن الـ homozygous للـ lethal allele يعيش لبعد الولادة ويموت بعد مدة من الزمن خلال فترة النمو
فهاي منسميها Delayed lethality ومثال عليها هو الـ Tay-Sachs disease و Huntington's & Friedreich ataxia
disease (بس Huntington's هو dominant lethal يعني اللي واحد فقط بسبب الوفاة بعمر كبير)

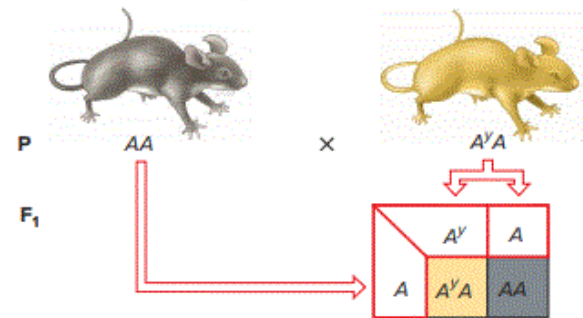
a. Tay-Sachs disease:

- normal newborns remain healthy for five to six months but then develop:
blindness, paralysis, mental impairment, and other symptoms of a deteriorating nervous system

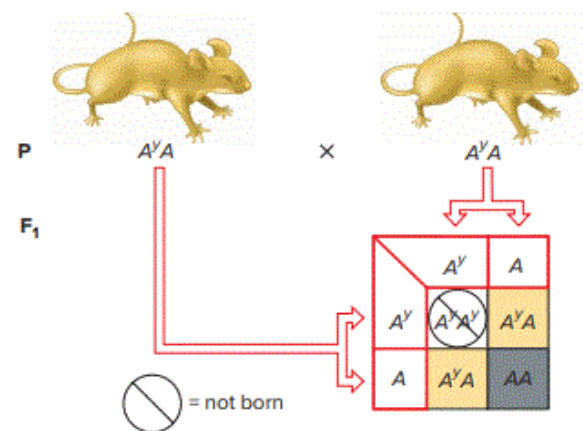
Figure 3.8 A^Y is a pleiotropic and recessive lethal allele.

(a) A cross between inbred agouti mice and yellow mice yields a 1:1 ratio of yellow to agouti progeny. The yellow mice are therefore $A^Y A$ heterozygotes, and for the trait of coat color, A^Y (for yellow) is dominant to A (for agouti). (Note we assume that if they could survive, $A^Y A^Y$ mice would have the same coat color as $A^Y A$ mice.)
(b) Yellow mice do not breed true. In a yellow x yellow cross, the 2:1 ratio of yellow to agouti progeny indicates that the A^Y allele is a recessive lethal.

(a) All yellow mice are heterozygotes.



(b) Two copies of A^Y cause lethality.



- Cause: absence of an active lysosomal enzyme called **hexosaminidase A**, leading to the accumulation of a toxic waste product inside nerve cells
 - ❖ Recessive alleles causing prenatal or early childhood lethality can be passed on to subsequent generations only by heterozygous carriers because affected homozygotes die before they can mate. (هاي النقطة كثير مهمة خلوها ببالكلم)
 - ❖ in late-onset disease that cause death in adults, homozygous patients can pass on the lethal allele before they become debilitated like in Friedreich ataxia and Huntington's disease
- b. The degenerative disease Friedreich ataxia:
- Some homozygotes first display symptoms of ataxia (loss of muscle coordination) at age 30–35 and die about five years later from heart failure.
- c. Huntington disease:
- Dominant lethal late-onset allele
 - can be transmitted to subsequent generations

نقطة مهمة هون لازم نحكيها وهي انو لو هاد ال dominant lethal allele ما كان late-onset وكان بسبب الموت prenataly فالشخص المصاب ما رح يوصل للعمر الي بسمحله ينقل جيناته للجيل الي بعده لهيك رح يختفي من ال population ولكن ممكن يظهر من جديد بسبب **new mutation**

SUMMARY:

TABLE 3.1 For Traits Determined by One Gene: Alterations of the 3:1 Monohybrid Ratio

What Mendel Described	Extension	Extension's Effect on Heterozygous Phenotype	Extension's Effect on Ratios Resulting from an $F_1 \times F_1$ Cross
Complete dominance	Incomplete dominance Codominance	Unlike either homozygote	Phenotypes coincide with genotypes in a ratio of 1:2:1
Two alleles	Multiple alleles	Multiplicity of phenotypes	A series of 3:1 or 1:2:1 ratios
All alleles are equally viable	Recessive lethal alleles	Heterozygotes survive but may have visible phenotypes	2:1 instead of 3:1
One gene determines one trait	Pleiotropy: One gene influences several traits	Several traits affected in different ways, depending on dominance relations	Different ratios, depending on dominance relations for each affected trait

هأ رح ناخذ مثال بجمعنا كل ال extensions الي اخدناها مع بعض :

A Comprehensive Example: Sickle-Cell Disease Illustrates Many Extensions to Mendel's View of Single-Gene Inheritance

Sickle-cell disease: a result of a faulty hemoglobin molecule

Hemoglobin is composed of two types of polypeptide chains:

- (1) alpha (α)-globin → Coded by *Hb α* gene
- (2) beta (β)-globin → Coded by *Hb β* gene

- Normal red blood cells are packed full of millions upon millions of hemoglobin molecules, each of which picks up oxygen in the lungs and transports it to all the body's tissues.

1st Extension: Multiple Alleles

- The β -globin gene has a normal wild-type allele ($Hb\beta^A$) that gives rise to fully functional β -globin, as well as close to 400 mutant alleles that have been identified so far
 - Some of these mutant alleles result in the production of hemoglobin that carries oxygen only inefficiently.
 - Other mutant alleles prevent the production of β -globin, causing a hemolytic (blood-destroying) disease called *β -thalassemia*.
- the most common mutant allele of the β -globin gene, $Hb\beta^S$, which specifies an abnormal polypeptide that causes sickling of red blood cells (Fig. 3.9a).

2nd Extension: Pleiotropy

- The $Hb\beta^S$ allele of the β -globin gene affects more than one trait (Fig. 3.9b)
 - Hemoglobin molecules in the red blood cells of homozygous $Hb\beta^S Hb\beta^S$ individuals undergo:
 - a. an aberrant transformation after releasing their oxygen.
 - Instead of remaining soluble in the cytoplasm, they aggregate to form long fibers that deform the red blood cell from a normal biconcave disk to a sickle shape (see Fig. 3.9a).
 - How does it affect?
 - (1) The deformed cells clog small blood vessels reducing oxygen flow to the tissues and giving rise to muscle cramps, shortness of breath, and fatigue.
 - (2) The sickled cells are also fragile and easily broken. Consumption of fragmented cells by phagocytic white blood cells leads to a low red blood cell count, a condition called *anemia*
 - b. Resistance to malaria because the organism that causes the disease, *Plasmodium falciparum*, can multiply rapidly in normal red blood cells but cannot do so in cells that sickle.
 - Why? because infection by *P. falciparum* causes sickle-shaped cells to break down before the malaria organism has a chance to multiply.

3rd Extension: Recessive lethality

People who are homozygous for the recessive $Hb\beta^S$ allele often develop heart failure because of stress on the circulatory system. Many sickle-cell sufferers die in childhood, adolescence, or early adulthood.

4th Extension: Different dominance relations

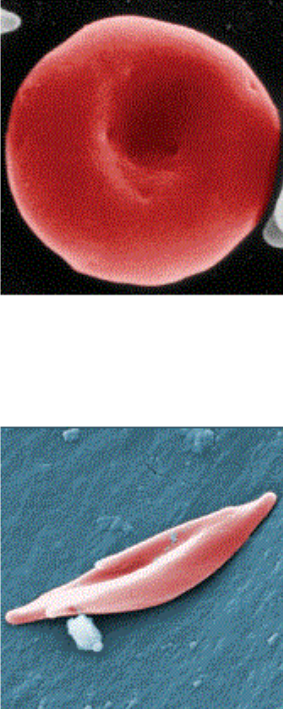
Comparisons of heterozygous carriers of the sickle-cell allele—individuals whose cells contain one $Hb\beta^A$ and one $Hb\beta^S$ allele—with homozygous $Hb\beta^A Hb\beta^A$ (normal) and homozygous $Hb\beta^S Hb\beta^S$ (diseased) individuals make it possible to distinguish different dominance relationships for different phenotypic aspects of sickle-cell anemia (Fig. 3.9b).

- At the Molecular level: both alleles are expressed such that $Hb\beta^A$ and $Hb\beta^S$ are codominant
- At the cellular level (effect on red blood cell shape): $Hb\beta^A$ and $Hb\beta^S$ alleles show either complete dominance or codominance depending on altitude
 - Under normal oxygen conditions, the great majority of a heterozygote's red blood cells have the normal biconcave shape ($Hb\beta^A$ is dominant to $Hb\beta^S$)
 - When oxygen levels drop, sickling occurs in some $Hb\beta^A Hb\beta^S$ cells ($Hb\beta^A$ and $Hb\beta^S$ are codominant)
- Considering the trait of resistance to malaria, the $Hb\beta^S$ allele is dominant to the $Hb\beta^A$ allele.

➔ Thus, for the β -globin gene, as for other genes, dominance and recessiveness are not an inherent quality of alleles in isolation; rather, they are specific to each pair of alleles and to the level of physiology at which the phenotype is examined.

Figure 3.9 Pleiotropy of sickle-cell anemia: Dominance relations vary with the phenotype under consideration. (a) A normal red blood cell (top) is easy to distinguish from the sickled cell in the scanning electron micrograph at the bottom. **(b)** Different levels of analysis identify various phenotypes. Dominance relationships between the $Hb\beta^S$ and $Hb\beta^A$ alleles of the $Hb\beta$ gene vary with the phenotype and sometimes even change with the environment.

a (top): © BSIP/Newscom; a (bottom): Source: Janice Haney Carr/CDC

	Phenotypes at Different Levels of Analysis	Normal $Hb\beta^A Hb\beta^A$	Carrier $Hb\beta^A Hb\beta^S$	Diseased $Hb\beta^S Hb\beta^S$	Dominance Relations at Each Level of Analysis
	Red blood cell shape at sea level	Normal	Normal	Sickled cells present	$Hb\beta^A$ is dominant $Hb\beta^S$ is recessive
	Red blood cell concentration at sea level	Normal	Normal	Lower	
	β -globin polypeptide production	A protein	A and S proteins	S proteins	$Hb\beta^A$ and $Hb\beta^S$ are codominant
	Red blood cell shape at high altitudes	Normal	Sickled cells present	Severe sickling	
	Red blood cell concentration at high altitudes	Normal	Lower	Very low, anemia	$Hb\beta^A$ and $Hb\beta^S$ show incomplete dominance
	Susceptibility to malaria	Normal susceptibility	Resistant	Resistant	$Hb\beta^S$ is dominant $Hb\beta^A$ is recessive

Concept: 3.2 Extensions to Mendel for Two-Gene Inheritance

Two genes can interact in several ways to determine a single trait, such as the color of a flower, a seed coat, a chicken's feathers, a dog's fur, or the shape of a plant's leaves.

- ➔ Novel phenotypes can result from gene interactions.
- ➔ Fewer phenotypes may be observed due to complementary gene action.
- ➔ Epistasis is when an allele at one gene masks the phenotype of alleles at another gene.
- ➔ Genes that perform the same function are **redundant**.

In all of these cases, F_2 phenotypes from dihybrid crosses vary from the expected 9:3:3:1 ratio.

أهم اشي بكل الامثلة الجاي انك تحفظ ال phenotypic ratios وكيف اختلفت عن ال 9:3:3:1 مع ال biochemical pathways لأنو هاد الاختلاف هو آلي رح يدلك بالامتحان على الحل وال biochemical pathways رح تساعدك بأنك تفسر سبب هاد الاختلاف.

Additive Interactions Between Two Genes Controlling a Single Trait Can Produce Novel Phenotypes

In the chapter opening, we described a mating of tan and gray lentils that produced a uniformly brown F_1 generation and then an F_2 generation containing brown, tan, gray, and green lentil seeds. An understanding of how this can happen emerges from experimental results demonstrating that the ratio of the four F_2 colors is 9 brown : 3 tan : 3 gray : 1 green (Fig. 3.10a)

(تذكروا انو حصلنا على هاي النسبة بالتشابتر الماضي مع مندل ولكن كانت نتيجة لتزواج صفتين مختلفتين مع بعض (dihybrid cross) زي مثلا البذور الصفراء الملساء والبذور الخضراء المجعدة ولكن هون عم نحكي عن صفة وحدة وهي لون البذور)

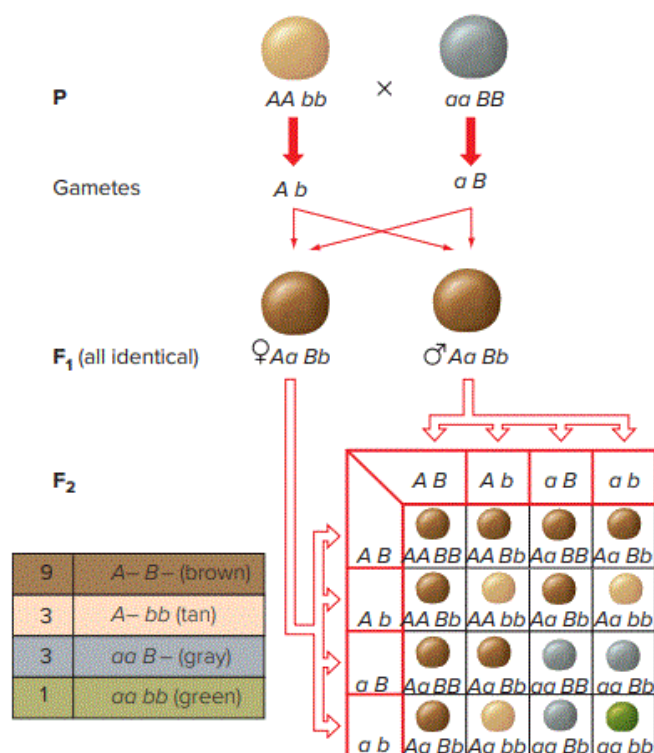
Explanation: a combination of genotypes at two independently assorting genes interacts additively to produce the phenotype of seed color in lentils

الاشي الي بدعم هاد التفسير هو ال **self-crossing** بين افراد ال F_2 .

- Self-crosses of F_2 green individuals show that they are pure-breeding, producing an F_3 generation that is entirely green.
 - Self-crosses of F_2 tan individuals generate either all tan offspring, or a mixture of tan offspring and green offspring.
 - Self-crosses of F_2 Gray individuals produce either all gray, or gray and green
 - Self-crosses of brown F_2 individuals can have four possible outcomes: all brown, brown plus tan, brown plus gray, or all four colors (Fig. 3.10b)
- ➔ The two-gene hypothesis explains why: •
- Only one green genotype exists: pure-breeding $aa\ bb$
 - Two types of tans exist: pure-breeding $AA\ bb$ as well as tan- and green-producing $Aa\ bb$
 - Two types of grays exist: pure-breeding $aa\ BB$ and gray- and green-producing $aa\ Bb$
 - Four types of browns exist: true-breeding $AA\ BB$, brown- and tan-producing $AA\ Bb$, brown- and gray producing $Aa\ BB$, and $Aa\ Bb$ dihybrids that give rise to plants producing lentils of all four colors

Figure 3.10 How two genes interact to produce seed colors in lentils. (a) In a cross of pure-breeding tan and gray lentils, all the F_1 hybrids are brown, but four different phenotypes appear among the F_2 progeny. The 9:3:3:1 ratio of F_2 phenotypes suggests that seed coat color is determined by two independently assorting genes. (b) Expected results of selfing individual F_2 plants of the indicated phenotypes to produce an F_3 generation, if seed coat color results from the interaction of two genes. The third column shows the proportion of the F_2 population that would be expected to produce the observed F_3 phenotypes. (c) Other two-generation crosses involving pure-breeding parental lines also support the two-gene hypothesis. In this table, the F_1 hybrid generation has been omitted.

(a) A dihybrid cross with lentil coat colors



(b) Self-pollination of the F_2 to produce an F_3

Phenotypes of F_2 Individuals	Observed F_3 Phenotypes	Expected Proportion of F_2 Population*
Green	Green	1/16
Tan	Tan	1/16
Tan	Tan, green	2/16
Gray	Gray, green	2/16
Gray	Gray	1/16
Brown	Brown	1/16
Brown	Brown, tan	2/16
Brown	Brown, gray	2/16
Brown	Brown, gray, tan, green	4/16

*This 1:1:2:2:1:1:2:2:4 F_2 genotypic ratio corresponds to a 9 brown : 3 tan : 3 gray : 1 green F_2 phenotypic ratio.

(c) Sorting out the dominance relations by select crosses

Seed Coat Color of Pure-Breeding Parents	F_2 Phenotypes and Frequencies	Ratio
Tan $\times$ green	231 tan, 85 green	3:1
Gray $\times$ green	2586 gray, 867 green	3:1
Brown $\times$ gray	964 brown, 312 gray	3:1
Brown $\times$ tan	255 brown, 76 tan	3:1
Brown $\times$ green	57 brown, 18 gray, 13 tan, 4 green	9:3:3:1

→ Summary:

- Two genes that determine seed color
 - both dominant alleles must be present to yield brown ($A- B-$);
 - the dominant allele of one gene produces tan ($A- bb$);
 - the dominant allele of the other specifies gray ($aa B-$); and
 - the complete absence of dominant alleles (that is, the double recessive) yields green ($aa bb$).
- Thus, the four color phenotypes arise from four genotypic classes, with each class defined in terms of the presence or absence of the dominant alleles of two genes:
 - both present ($A- B-$)
 - one present ($A- bb$)
 - the other present ($aa B-$)
 - neither present ($aa bb$)

→ Biochemical explanation:

(the genes controlling lentil seed color have not been identified at the molecular level, and the biochemical pathways in which they function are not known. However, information available about the mechanisms of seed color inheritance in other plant species allows us to formulate a plausible model for this system in lentils (Fig. 3.11))

- Gene A- code for tan color enzyme that changes color precursor 1 to tan color
- Gene B- code for gray color enzyme that changes color precursor 2 to gray color

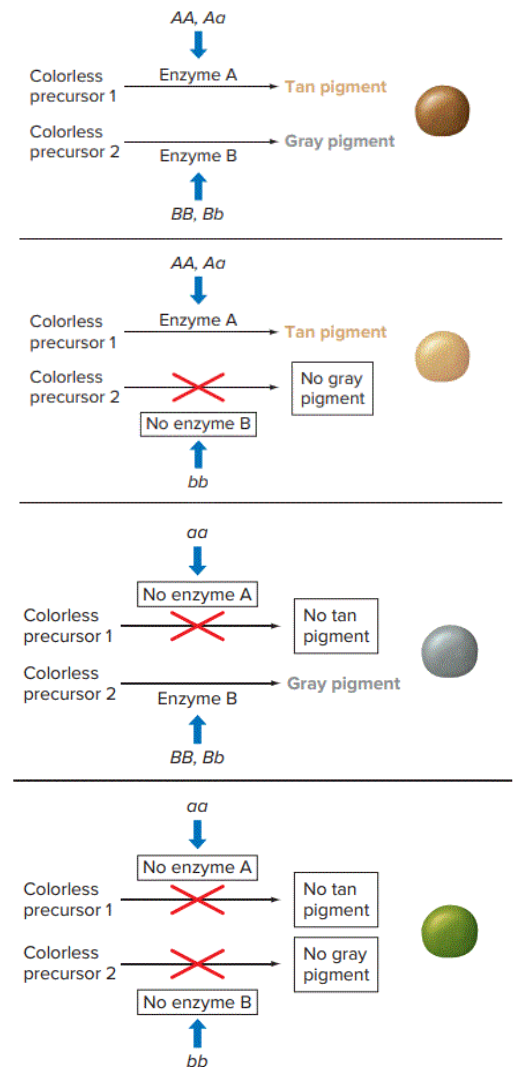
→ gene A- + gene B- = Enzyme A + Enzyme B = Brown color (tan + gray)

→ Gene A- + No gene B (bb) = enzyme A only = Tan color

→ No gene A (aa) + Gene B- = Enzyme B only = Gray color

→ No A (aa) + No B (bb) = no tan and gray enzymes = Green color

Figure 3.11 A biochemical model for the inheritance of lentil seed colors. The seed has an opaque outer layer (the seed coat) and an inner layer (the cotyledon). The green chlorophyll in the cotyledon is not visible if the seed coat is colored. Allele A encodes enzyme A. Allele a does not produce this enzyme. Allele B of a second gene encodes a different enzyme; b produces none of this enzyme. Seeds appear brown if the tan and gray pigments are both present. In the absence of both enzymes ($aa bb$), the seed coat is nonpigmented, so the green chlorophyll in the cotyledon will show through. The 9:3:3:1 ratio implies that the A and B genes operate in independent biochemical pathways.



Epistasis: Alleles of One Gene Can Mask the Phenotypic Effects of Alleles of Another Gene

Epistasis: A gene interaction in which an allele at one gene hides the effects of alleles at another gene

→ the allele that is doing the masking is *epistatic* to the gene that is being masked (the hypostatic gene).

A. Recessive epistasis: Homozygosity for a recessive allele of one gene hides the effect of a second gene.

Examples:

1) Yellow Labrador retrievers:

The sleek, short-haired coat of Labrador retrievers can be black, chocolate brown, or yellow (Fig. 3.12a)

→ Which color appears depends on the allelic combinations of two independently assorting coat color genes (Fig. 3.12b).

- Crosses between pure-breeding black retrievers ($BB EE$) and one type of pure-breeding yellow retriever ($bb ee$) create an F_1 generation of dihybrid black retrievers ($Bb Ee$).

→ Crosses between these F_1 dihybrids produce an F_2 generation with nine black dogs ($B- E-$) for every three brown ($bb E-$) and four yellow ($- - ee$) (Fig. 3.12b)

❖ only three phenotypic classes exist because the two genotypic classes without a dominant E allele—the three $B- ee$ and the one $bb ee$ —combine to produce a yellow phenotype

- **Ratio:** recessive epistasis ratio in the F_2 generation is **9:3:4**, with the 4 representing a combination of 3 ($B- ee$) + 1 ($bb ee$).

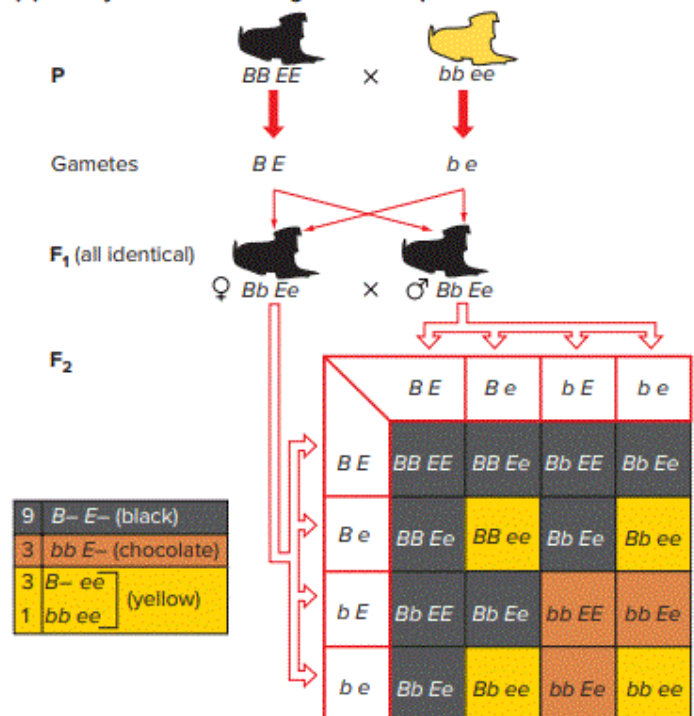
Figure 3.12 Recessive epistasis determines coat color in Labrador retrievers. (a) Labrador retriever colors. (b) Yellow Labrador retrievers are homozygous for the recessive e allele, which masks the effects of the B or b alleles of a second coat color gene. In $E-$ dogs, a $B-$ genotype produces black and a bb genotype produces brown.

a: © Vanessa Grosemy/Alamy

(a) Chocolate, yellow, and black Labrador retrievers



(b) A dihybrid cross showing recessive epistasis



- **Biochemical explanation (Fig3.13):** All coat color in dogs comes from two pigments synthesized from a common precursor:

1. a dark pigment called eumelanin
2. a light pigment called pheomelanin

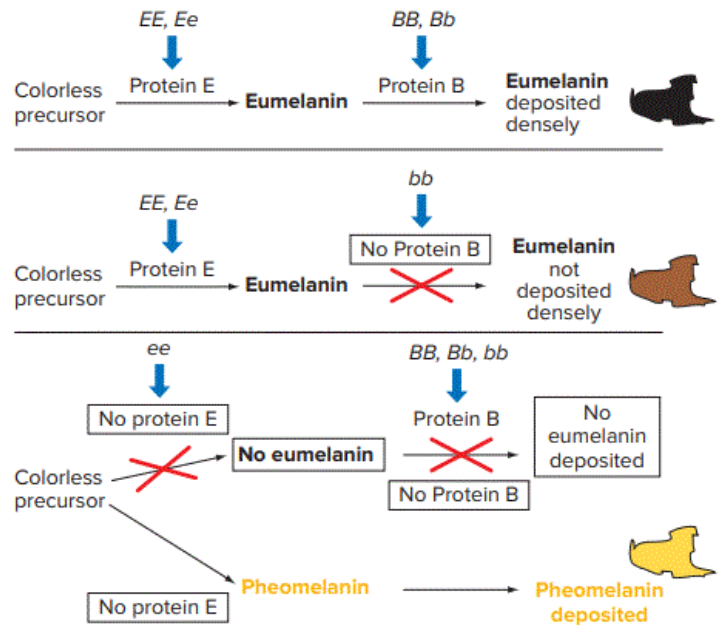
→ When Labrador retrievers have at least one copy of the E allele, the resultant protein E ensures that the animals will make only eumelanin and no pheomelanin.

- protein specified by the B allele is required for eumelanin synthesis and its deposition in the hair, while the protein made by the b allele is less efficient. As a result, chocolate E- bb dogs have less eumelanin in their hairs than black dogs with at least one B allele (E- B-).
- in the absence of the E protein (in ee dogs), only pheomelanin is synthesized, and so the dogs appear yellow

→ Why ee is epistatic to both alleles of gene B?

In ee dogs, no eumelanin is present, so the dogs are yellow regardless of whether they are B- or bb.

Figure 3.13 A biochemical explanation for coat color in Labrador retrievers. Protein E activates an enzyme that generates eumelanin from a colorless precursor. When protein E is present, only eumelanin is produced. Protein B deposits eumelanin densely so that the hair is black. Pigment is deposited less densely without protein B, producing brown hair (chocolate). In the absence of protein E, no eumelanin is produced and instead, pheomelanin (yellow pigment) is synthesized. Homozygous ee dogs are always yellow regardless of the gene B genotype. The reason is that protein B affects eumelanin only, but these ee animals have no eumelanin.



2) The Bombay phenotype in humans:

In rare instances, two parents who appear to have blood type O, and thus would be predicted to be genotype ii, produce a child who is either blood type A (genotype I^A i) or blood type B (genotype I^B i)

(called the Bombay phenotype after its discovery in Bombay, India, superficially resembles blood type O)

- the Bombay phenotype arises from homozygosity for a mutant recessive allele (hh) of a second gene that masks the effects of any ABO alleles that might be present. (Fig. 3.14a)

→ How it works at the molecular level? (Fig. 3.14a)

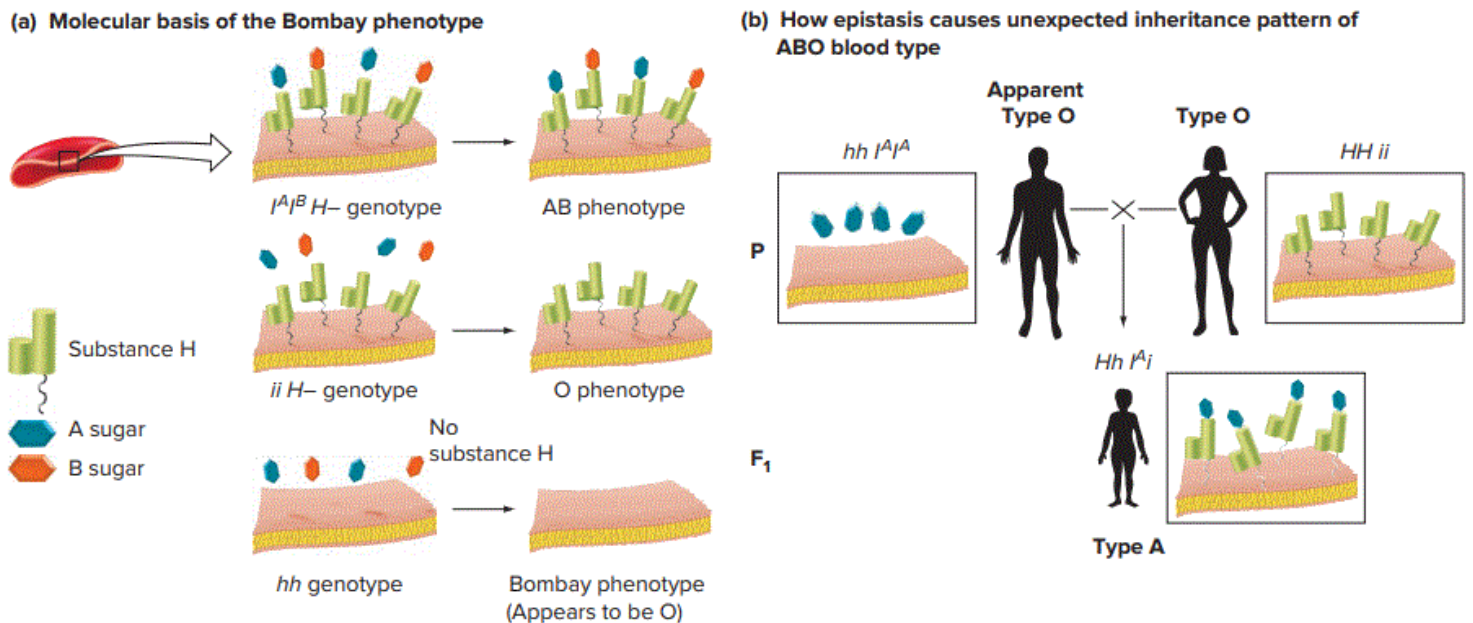
In the construction of the red blood cell surface molecules that determine blood type, it is found that:

- type A individuals make an enzyme that adds polysaccharide A onto a sugar polymer known as substance H
- type B individuals make an altered form of the enzyme that adds polysaccharide B onto the sugar polymer H

- type O individuals make neither A-adding nor B-adding enzyme and thus have an exposed substance H in the membranes of their red blood cells.
 - All people of A, B, or O phenotype carry at least one dominant wild-type H allele for the second gene and thus produce some substance H.
- ➔ In contrast the rare Bombay-phenotype individuals, with genotype hh for the second gene, do not make substance H at all.
- ➔ Thus, even if these people make an enzyme that would add A or B to this polysaccharide base, they have nothing to add it onto; as a result, Bombay-phenotype individuals appear to be type O. For this reason, homozygosity for the recessive h allele of the H-substance gene masks the effects of the ABO gene, making the hh genotype epistatic to any combination of I^A , I^B , and i alleles (except for ii .)

Fig. 3.14b: A person who carries I^A , I^B , or both I^A and I^B but is also an hh homozygote for the H-substance gene may appear to be type O, but he or she will be able to pass along an I^A or I^B allele in sperm or egg. An offspring receiving, let's say, an I^A allele for the ABO gene and a recessive h allele for the H-substance gene from the father plus an i allele and a dominant H allele from the mother would have blood type A (genotype $I^A i Hh$), even though neither of the parents is phenotype A or AB.

Figure 3.14 Recessive epistasis in humans causes a rare blood type. (a) Homozygosity for the h Bombay allele is epistatic to the I gene determining ABO blood types. hh individuals fail to produce substance H, which is needed for the addition of A or B sugars at the surface of red blood cells. (b) Because h is epistatic to I , rare individuals may appear to have blood type O despite having an I^A or I^B allele. When the masked I allele is expressed in their Hh progeny, these people may be surprised by their child's blood type.



3) White sweet pea flowers (**Reciprocal recessive epistasis**)

In the first decade of the twentieth century, William Bateson conducted a cross between two lines of pure-breeding white-flowered sweet peas (Fig. 3.15a)

→ Quite unexpectedly, all of the F_1 progeny were purple! (Fig. 3.15b).

- Self-pollination of these novel hybrids produced a **ratio of 9 purple: 7 white** in the F_2 generation.

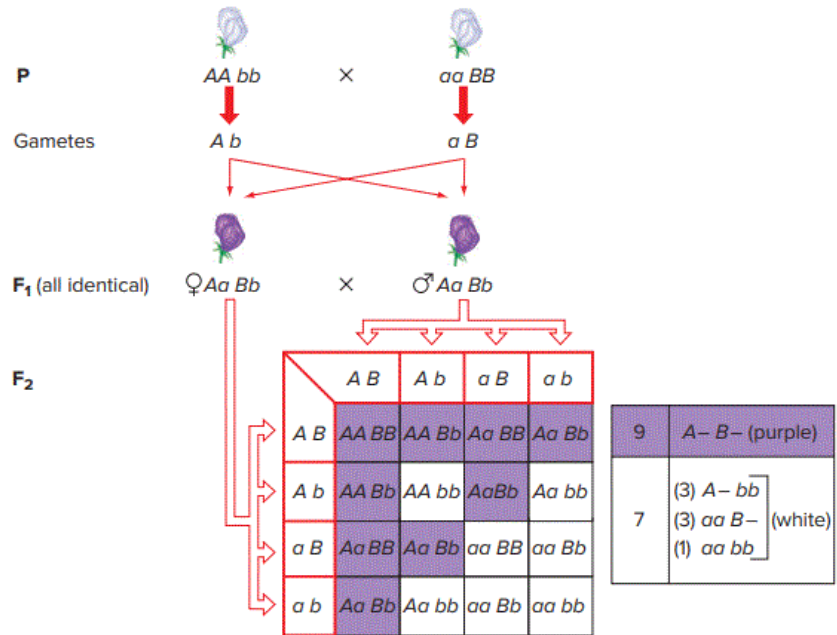
Figure 3.15 Dominant alleles of two genes needed for purple color in sweet peas. (a) White and purple sweet pea flowers. (b) The 9:7 ratio of purple to white F_2 plants indicates that at least one dominant allele of each gene is necessary for the development of purple color.

© William Allen/National Geographic Creative

(a) *Lathyrus odoratus* (sweet peas)



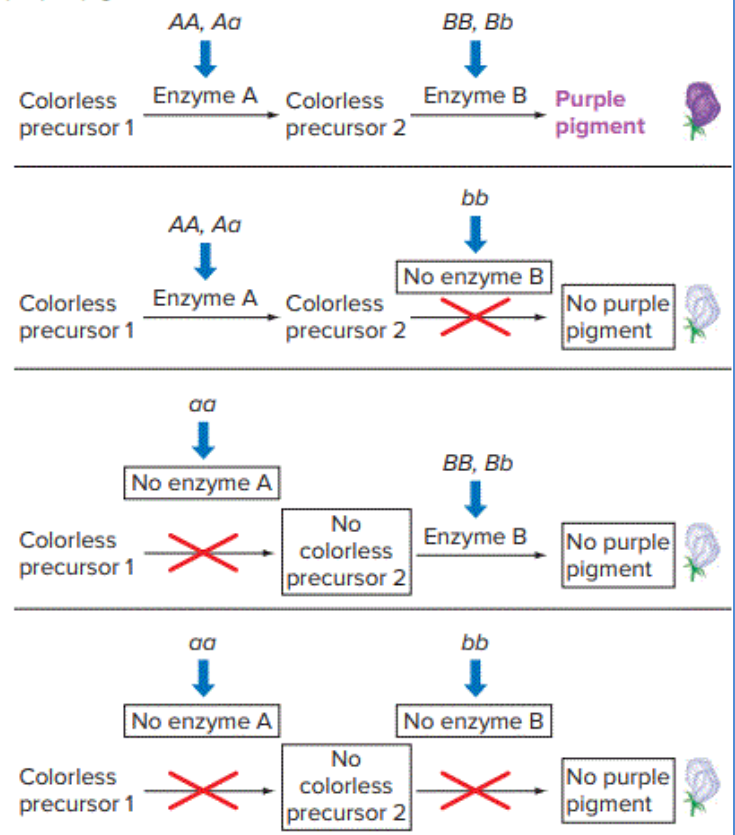
(b) A dihybrid cross showing reciprocal recessive epistasis



- **Explanation:** Two genes work in tandem to produce purple sweet pea flowers, and a dominant allele of each gene must be present to produce that color.
- **Biochemical hypothesis (Fig. 3.16):**
 - it takes two enzymes to catalyze sequential biochemical reactions in order to change a colorless precursor into a purple pigment
 - Only the $A- B-$ genotypic class, which produces active forms of both required enzymes, can generate colored flowers.
 - The other three genotypic classes ($A- bb$, $aa B-$, and $aa bb$) become grouped together with respect to phenotype **because they do not specify functional forms of one or the other requisite enzyme** and thus give rise to no color, which is the same as white.

→ It is easy to see how the 7 part of the 9:7 ratio encompasses the 3:3:1 of the 9:3:3:1 F_2 ratio.

Figure 3.16 A biochemical explanation for reciprocal recessive epistasis in the generation of sweet pea color. Enzymes specified by the dominant alleles of two genes are both necessary to produce pigment. The recessive alleles of both genes specify no enzymes. In aa homozygotes, no intermediate precursor 2 is generated, so even if enzyme B is available, it cannot produce purple pigment.



- ❖ The 9:7 ratio is the phenotypic signature of this type of **reciprocal recessive epistasis** in which the dominant alleles of two genes acting together ($A- B-$) produce color or some other trait, while the other three genotypic classes ($A- bb$, $aa B-$, and $aa bb$) do not (see Fig. 3.15b).
- ❖ Given that the phenotype associated with either allele A or allele B is purple, then we can say that aa is epistatic to B , and bb is epistatic to A . If the sweet peas are either aa or bb , their flowers will be white regardless of whether or not they have a dominant allele of the other gene.

❖

B. Dominant Epistasis: Epistasis in which the dominant allele of one gene hides the effects of another gene

- **Examples:**

1) Squash fruit color:

In summer squash, two genes influence the color of the fruit (Fig. 3.17a):

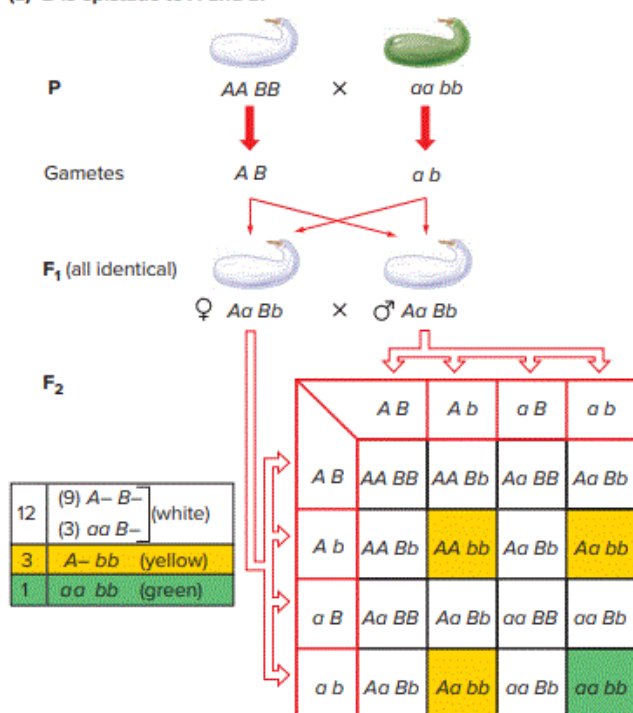
- In the first gene, the dominant allele ($A-$) determines yellow, while homozygotes for the recessive allele (aa) are green.
- A second gene's dominant allele ($B-$) produces white, while bb fruit may be either yellow or green, depending on the genotype of the first gene

→ In the interaction between these two genes, the presence of B hides the effects of either $A-$ or aa , producing white fruit, and $B-$ is thus epistatic to any genotype of the A gene. The recessive b allele has no effect on fruit color determined by gene A .

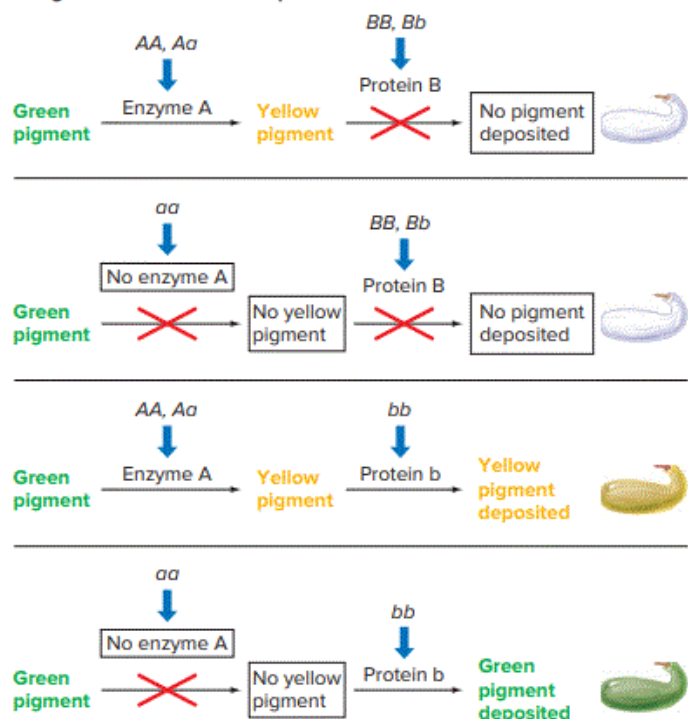
- In a cross between white F_1 dihybrids ($Aa Bb$), the F_2 phenotypic ratio is **12 white : 3 yellow : 1 green** (Fig. 3.17a). The 12 includes two genotypic classes: 9 $A- B-$ and 3 $aa B-$.

Figure 3.17 Dominant epistasis may result in a 12:3:1 phenotypic ratio. (a) In summer squash, the dominant B allele causes white color and is sufficient to mask the effects of any combination of A and a alleles. As a result, yellow ($A-$) or green (aa) color is expressed only in bb individuals. (b) The A allele encodes enzyme A , while the a allele specifies no enzyme. Therefore, yellow pigment is present in $A-$ squash and green pigment in aa squash. Deposition of either pigment depends on protein b encoded by allele b , the normal (wild-type) allele of a second gene. However, the mutant dominant allele B encodes an abnormal version B of this protein that prevents pigment deposition, even when the normal protein b is present. Therefore, in order to be colored, the squash must have protein b but not protein B (genotype bb).

(a) B is epistatic to A and a .



(b) A possible biochemical explanation for dominant epistasis in the generation of summer squash color



- **Biochemical hypothesis:** The squash genes A and B have not been identified at the molecular level, and the biochemical pathway in which they interact is unknown. However, based on knowledge of similar phenomena in other plants, a likely biochemical pathway underlying the 12:3:1 phenotypic ratio is shown in Fig. 3.17b.

2) Chicken feather color:

White leghorns have a doubly dominant AA BB genotype for feather color; white wyandottes are homozygous recessive for both genes (aa bb)

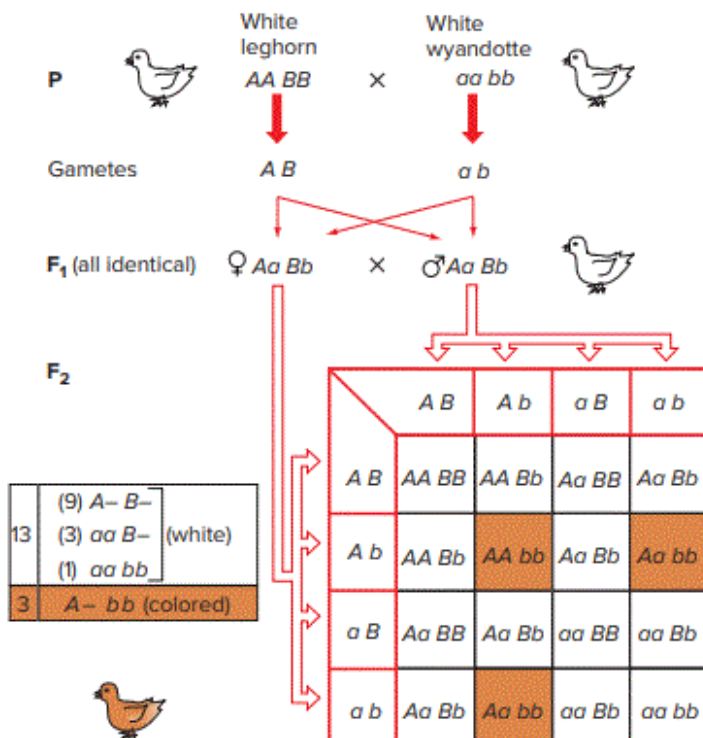
→ A cross between these two pure-breeding white strains produces an all-white dihybrid (Aa Bb) F₁ generation but birds with color in their feathers appear in the F₂

❖ **ratio of white to colorful is 13:3** (Fig. 3.18a).

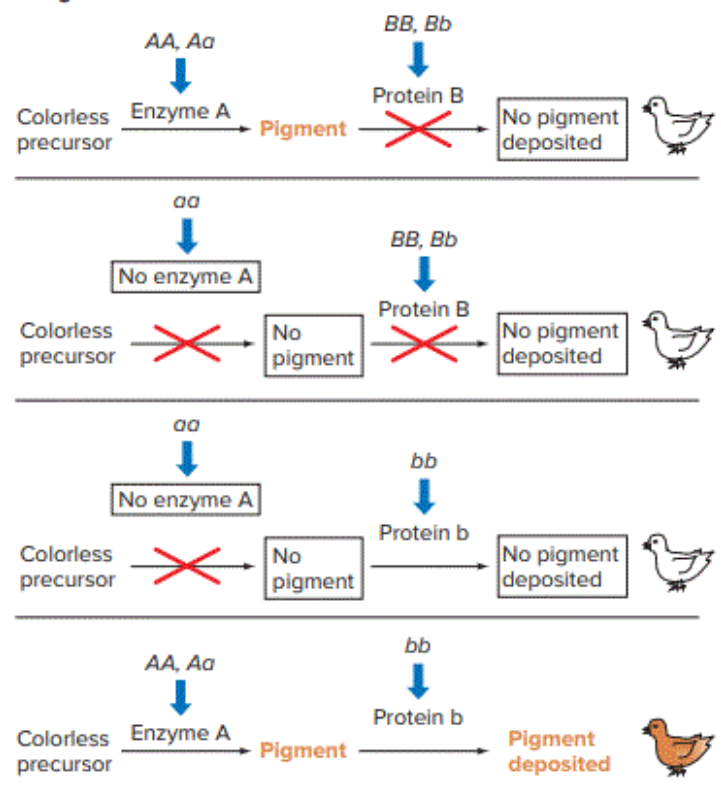
- **Explanation:** a kind of dominant epistasis in which B is epistatic to A
→ the A allele produces color only in the absence of B; and the a, B, and b alleles produce no color. The interaction is characterized by a 13:3 ratio because the 9 A– B–, 3 aa B–, and 1 aa bb genotypic classes combine to produce only one phenotype: white
- **Biochemical explanation:** The biochemical pathway known to underly the 13:3 ratio for chicken feather color is shown in Fig. 3.18b.

Figure 3.18 Dominant epistasis may also result in a 13:3 phenotypic ratio. (a) In the F₂ generation resulting from a dihybrid cross between white leghorn and white wyandotte chickens, the ratio of white birds to birds with color is 13:3. This ratio emerges because at least one copy of A and the absence of B is needed to produce color. **(b)** Enzyme A, encoded by allele A, is needed to synthesize pigment. Allele a encodes no enzyme. Pigment deposition in the feathers depends on protein b encoded by allele b, the normal (wild-type) allele of a second gene. The mutant dominant allele B, however, encodes an abnormal version of the protein that prevents pigment deposition, even when the normal protein b is present.

(a) B is epistatic to A



(b) Biochemical explanation for dominant epistasis in the generation of chicken feather color



- **Important points regarding epistasis:**

- Epistasis is an interaction between alleles of different genes, not between alleles of the same gene.
- In dihybrid crosses, the F₂ phenotypic ratios resulting from epistasis depend on the functions of the specific alleles and the particular biochemical pathways in which the genes participate
- Recessive epistasis usually indicates that the dominant alleles of the two genes function in the same pathway to achieve a common outcome. In the Labrador retriever example, B and E both function to generate black hairs.
- Dominant epistasis usually indicates that the dominant alleles of the two genes have antagonistic functions. Both in the cases of squash and chicken color, the dominant allele of gene B prevents deposition of a pigment whose synthesis depends on the dominant allele of gene A.

Redundancy: One or More Genes in a Pathway Are Superfluous

In maize, two genes, A and B, control leaf development

- Normal broad leaves develop as long as the plant has either a dominant A allele or a dominant B allele (A- B-, A- bb, or aa B-).
- the leaves of plants that have neither dominant allele (aa bb) are skinny because they contain too few cells (Fig. 3.19a)
- Given that leaves are malformed only in the absence of both A and B (aa bb), the F₂ phenotypic ratio signifying redundant gene action is **15:1** (Fig. 3.19b).

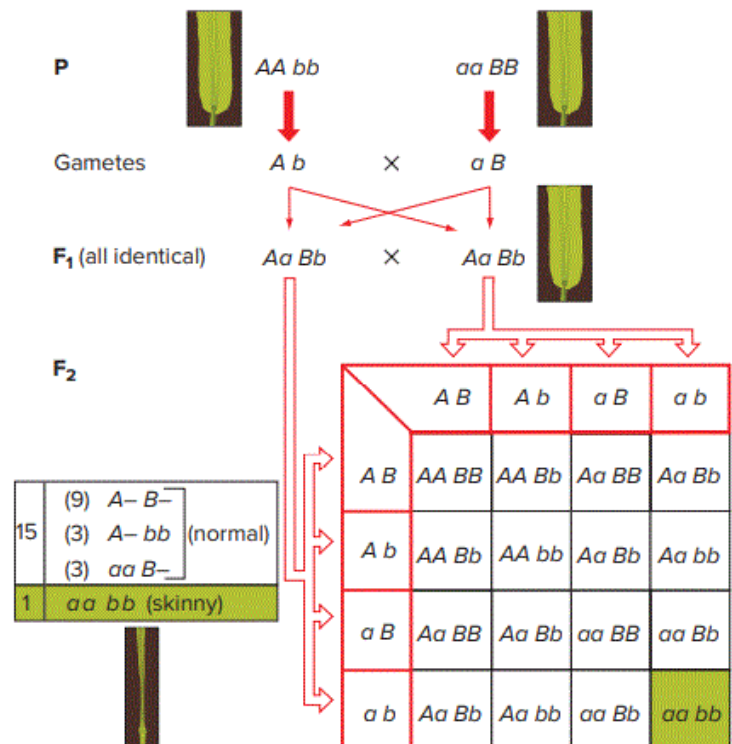
Figure 3.19 Redundant genes result in a 15:1 phenotypic ratio. (a) Normal maize leaves (AA BB) and a leaf lacking both dominant alleles A and B (aa bb). **(b)** In maize, either dominant allele A or B is sufficient for normal leaf development. Only the absence of both dominant alleles (aa bb) results in malformed, thin leaves. The result is a 15:1 ratio in the dihybrid cross F₂.

a: © Dr. Michael J. Scanlon, Cornell University

(a) Normal maize leaves and mutants lacking two redundant gene functions

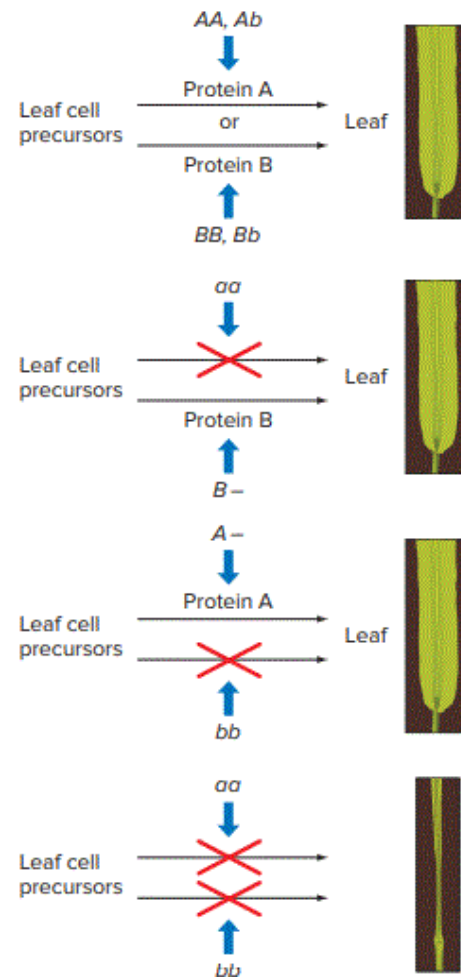


(b) A and B are redundant for maize leaf development



- **Biochemical explanation:** The proteins (A and B) encoded by the dominant alleles act in parallel, redundant pathways that recruit precursor cells to become part of the leaf (Fig. 3.20). That is, if either pathway functions, the leaves will develop their normal broad shape. Often, as in this case, redundant genes specify nearly identical proteins that perform the same function. Why does the organism have two genes that do the same thing? One answer is that redundant genes often arise by chance evolutionary processes that duplicate genes

Figure 3.20 A biochemical explanation for redundant gene action. The dominant alleles *A* and *B* specify proteins that function in independent pathways to instruct cells to become part of the leaf. The recessive alleles *a* and *b* specify no proteins. Because either pathway is sufficient, only plants that lack both dominant alleles have thin leaves.



SUMMARY:

TABLE 3.2 Summary of Two-Gene Interactions

Gene Interaction	Example	F ₂ Genotypic Ratios from an F ₁ Dihybrid Cross				F ₂ Phenotypic Ratio
		A- B-	A- bb	aa B-	aa bb	
Additive: Four distinct F ₂ phenotypes	Lentil: seed coat color (see Fig. 3.10a)	9	3	3	1	9:3:3:1
Recessive epistasis: When homozygous, recessive allele of one gene masks both alleles of another gene	Labrador retriever: coat color (see Fig. 3.12b)	9	3	3	1	9:3:4
Reciprocal recessive epistasis: When homozygous, recessive allele of each gene masks the dominant allele of the other gene	Sweet pea: flower color (see Fig. 3.15b)	9	3	3	1	9:7
Dominant epistasis I: Dominant allele of one gene hides effects of both alleles of the other gene	Summer squash: color (see Fig. 3.17a)	9	3	3	1	12:3:1
Dominant epistasis II: Dominant allele of one gene hides effects of dominant allele of other gene	Chicken feathers: color (see Fig. 3.18a)	9	3	3	1	13:3
Redundancy: Only one dominant allele of either of two genes is necessary to produce phenotype	Maize: leaf development (see Fig. 3.19b)	9	3	3	1	15:1

Incomplete Dominance or Codominance Can Expand Phenotypic Variation

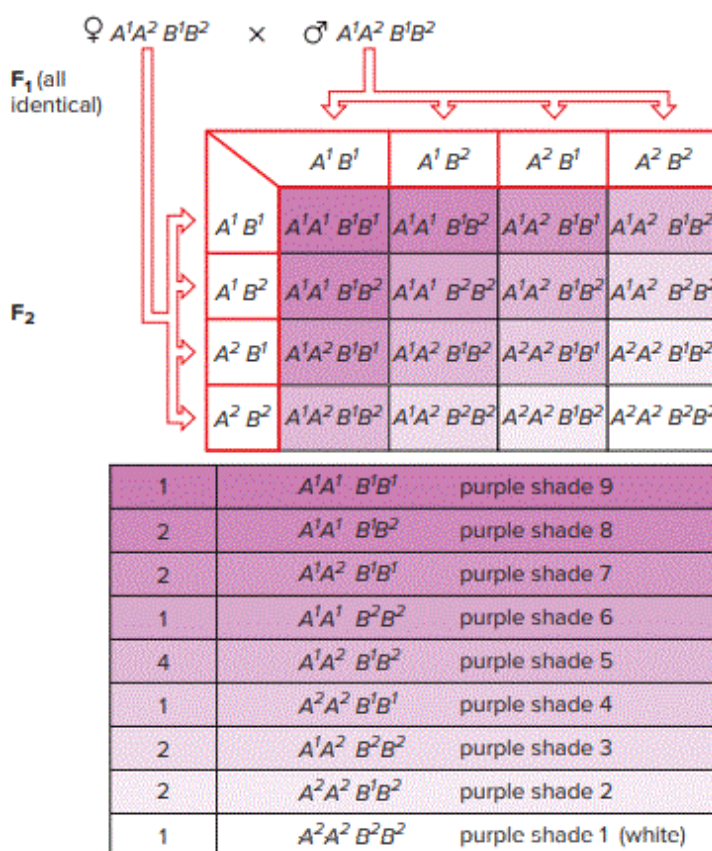
for any type of gene interaction, the alleles of one or both genes may exhibit incomplete dominance or codominance, and these possibilities increase the potential for phenotypic diversity.

- Example: Fig. 3.21 shows how incomplete dominance at both genes in a dihybrid cross results not in a collapse of several genotypic classes into one but rather an expansion—each of the nine genotypes in the dihybrid cross F_2 corresponds to a different phenotype.

بالأمثلة السابقة كانت كل مجموعة من الـ genotype نحطهم مع بعض لأنهم ينتجوا phenotype واحد (زي مثلا بالـ redundancy phenotype). $A-B-$, $A-bb$, and $aaB-$ incomplete dominance او complete phenotype. لجينين ينتج عنا phenotype مختلف لكل genotype لهيك بصير في تنوع بكل phenotype حسب شو الـ genotype تبعه.

- Biochemical explanation: purple pigmentation requires the action of **two enzymes**, A and B, and one is more efficient than the other, resulting in one gene (in this case, the A gene) contributing more to the purple phenotype than the other gene. (similar to that for incomplete dominance in Fig. 3.3b, where the amount of red pigment produced was proportional to the amount of **an enzyme**)

Figure 3.21 With incomplete dominance, the interaction of two genes can produce nine different phenotypes for a single trait. In this example, two genes produce purple pigments. Alleles A^1 and A^2 of the first gene exhibit incomplete dominance, as do alleles B^1 and B^2 of the second gene. The two alleles of each gene can generate three different phenotypes, so double heterozygotes can produce nine (3×3) different colors in a ratio of 1:2:2:1:4:1:2:2:1.



Breeding Studies Help Geneticists Determine Whether One or Two Genes Determine a Trait

How do geneticists know whether a particular trait is caused by the alleles of one gene or by two genes interacting in one of a number of possible ways?

Breeding tests can usually resolve the issue. Phenotypic ratios diagnostic of a particular mode of inheritance (for instance, the 9:7 or 13:3 ratios indicating that two genes are interacting) can provide the first clues and suggest hypotheses. Further breeding studies can then show which hypothesis is correct.

- Examples: a mating of one strain of pure-breeding white albino mice with pure-breeding brown results in black hybrids; and a cross between the black F_1 hybrids produces 90 black, 30 brown, and 40 albino offspring.

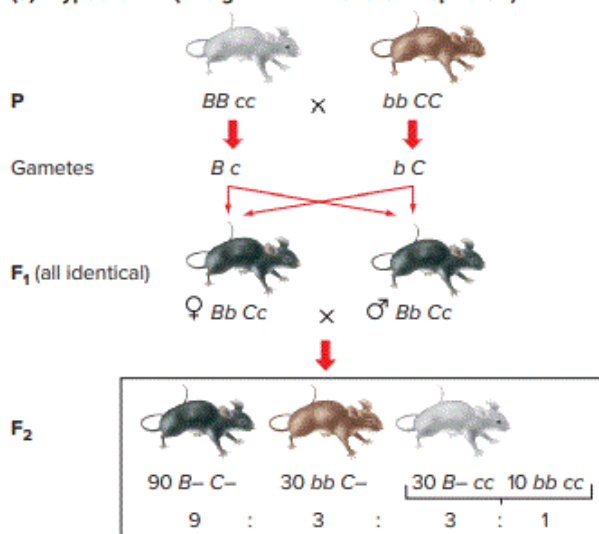
→ What is the genetic constitution of these phenotypes?

We can draw two hypotheses:

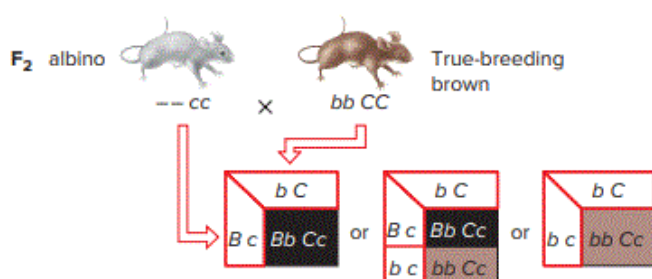
- (1) We could assume that we are seeing the 9:3:4 ratio of recessive epistasis and hypothesize that two interacting genes (call them B and C) control color. In this model, each gene has completely dominant and recessive alleles, and the homozygous recessive of one gene is epistatic to both alleles of the other gene (Fig. 3.22a)
- If two independently assorting genes and recessive epistasis, then F₂ progeny should be in 9:3:4 ratio
- (2) We might also explain the data—160 progeny in a ratio of 90:30:40—by the activity of one gene (Fig. 3.22b). According to this one-gene hypothesis, albinos would be homozygotes for one allele (B¹ B¹), brown mice would be homozygotes for a second allele (B² B²), and black mice would be heterozygotes (B¹ B²) that have their own novel phenotype because B¹ and B² are incompletely dominant. Under this system, a mating of black (B¹ B²) to black (B¹ B²) would be expected to produce 1 B² B² brown : 2 B¹ B¹ black : 1 B¹ B¹ albino, or 40 brown : 80 black : 40 albino
- If single gene with incomplete dominance, then F₂ progeny should be in 1:2:1 ratio
- **These ratios can appear very similar. Further breeding studies can reveal which hypothesis is correct.**

Figure 3.22 Specific breeding tests can help decide between hypotheses. Either of two models could explain the results of a cross tracking coat color in mice. **(a)** In one hypothesis, two genes interact with recessive epistasis to produce a 9:3:4 ratio. **(b)** In the other hypothesis, a single gene with incomplete dominance between the alleles generates the observed results. One way to decide between these models is to cross each of several albino F₂ mice with true-breeding brown mice. The two-gene model predicts several different outcomes depending on the *cc* albino's genotype at the *B* gene. The one-gene model predicts that all progeny of all the crosses will be black.

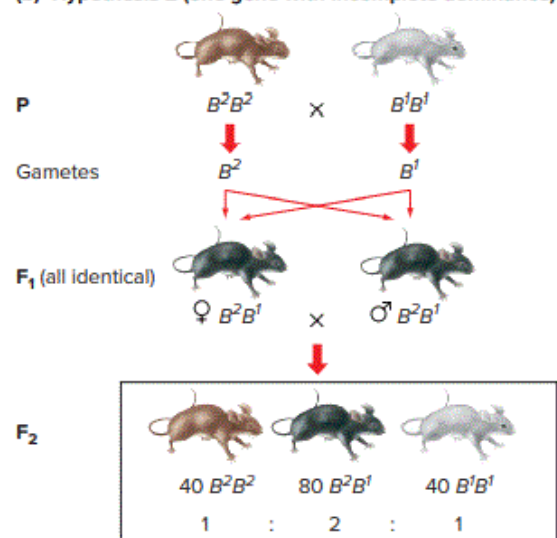
(a) Hypothesis 1 (two genes with recessive epistasis)



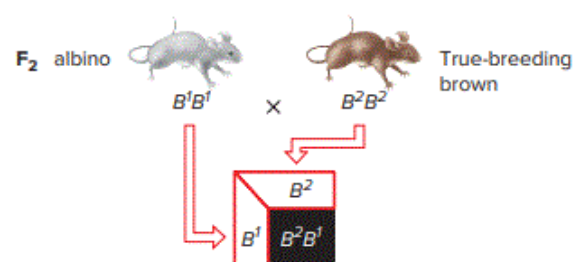
If two-gene hypothesis is correct:



(b) Hypothesis 2 (one gene with incomplete dominance)



If one-gene hypothesis is correct:



- ☞ if the one gene hypothesis were correct, a mating of F_2 albinos with pure-breeding brown mice similar to those of the parental generation would produce all black heterozygotes [brown (BB) $\times$ albino (bb) = all black (Bb)] (Fig. 3.22b)
- ☞ But if the two-gene hypothesis is correct, with recessive mutations at an albino gene (called C) epistatic to all expression from the B gene, different matings of pure-breeding brown (bb CC) with the F_2 albinos (— cc) will give different results—all progeny are black; half are black and half brown; all are brown—depending on the albino's genotype at the B gene (see Fig. 3.22a).
 - ➔ In fact, when the experiment is actually performed, the diversity of results confirms the two-gene hypothesis.

Locus Heterogeneity: Mutations in Any One of Several Genes May Cause the Same Phenotype

Locus Heterogeneity: A mutation at any one of a number of genes can give rise to the same phenotype.

- Close to 50 different genes have mutant alleles that can cause deafness in humans
 - ➔ Many genes generate the developmental pathway that brings about hearing, and a loss of function in any part of the pathway, for instance, in one small bone of the middle ear, can result in deafness (In other words, it takes a dominant wild-type allele at each of these 50 genes to produce normal hearing)

• Evidence for locus heterogeneity in human pedigrees

Example:

➔ whether a particular nonhearing man and a particular nonhearing woman carry mutations in the same gene or different ones can be determined if they have children together. If they have only children who can hear, the parents most likely carry mutations at two different genes, and the children carry one normal, wild-type allele for both of those genes (Fig. 3.23a).

➔ By contrast, if all of their children are deaf, it is likely that both parents are homozygous for a mutation in the same gene, and all of their children are also homozygous for this same mutation (Fig. 3.23b).

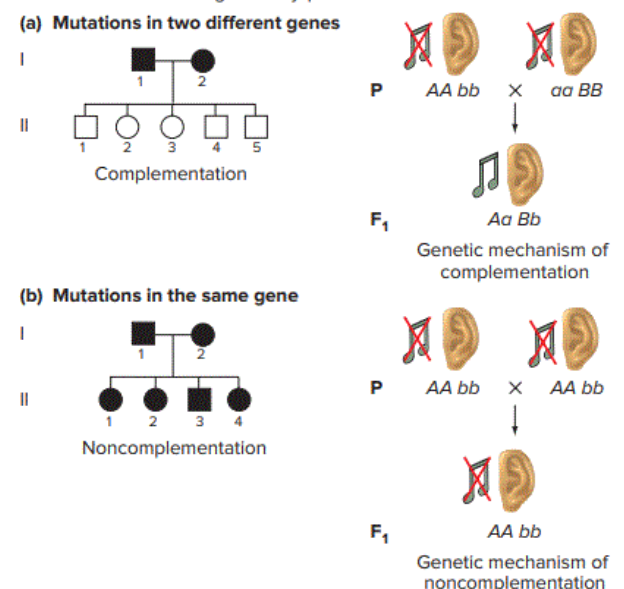
• Complementation and complementation tests

A test used to determine if a particular phenotype arises from mutations in the same or separate genes
➔ Can be applied only with recessive, not dominant, phenotypes

- If offspring receiving the two mutations—one from each parent—express the wild-type phenotype, Then ➔ **complementation** has occurred.

The observation of complementation means that:
the original mutations affected two different genes,
and for both genes, the normal allele from one parent can provide what the mutant allele of the same gene from the other parent cannot.

Figure 3.23 Locus heterogeneity in humans: Mutations in many genes can cause deafness. (a) Two deaf parents can have hearing offspring if the mother and father are homozygous for recessive mutations in different genes. (b) Two deaf parents with mutations in the same gene may produce all deaf children.



- Examples on locus heterogeneity:
 1. sweet pea flowers (seen in figure 3.15)
 2. Deafness: (Figure 3.23)
 3. Ocular-cutaneous albinism (OCA) (Figure 3.24):
 - People with this inherited condition have little or no pigment in their skin, hair, and eyes (Fig. 3.24a)
 - The horizontal inheritance pattern seen in Fig. 3.24b suggests that OCA is determined by the recessive allele of one gene, with albino family members being homozygotes for that allele
 → But a 1952 paper on albinism reported a family in which two albino parents produced three normally pigmented children (Fig. 3.24c). How would you explain this phenomenon?

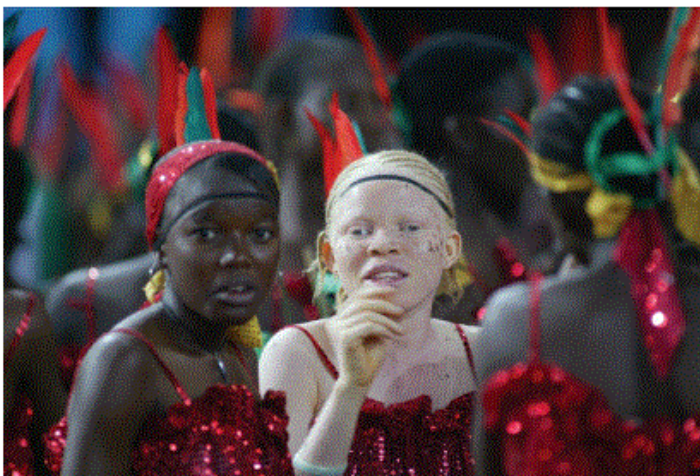
The answer is that albinism is another example of locus heterogeneity: Mutant alleles at any one of several different genes can cause the condition.

The complementation observed showed that one albino parent was homozygous for an OCA-causing mutation in gene A, while the other albino parent was homozygous for an OCA-causing mutation in a different gene, B.

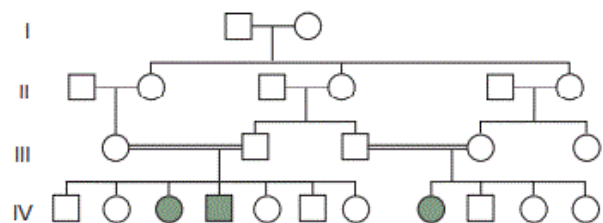
Figure 3.24 Family pedigrees help unravel the genetic basis of ocular-cutaneous albinism (OCA). (a) An albino Nigerian girl and her sister celebrating the conclusion of the All Africa games. (b) A pedigree following the inheritance of OCA in an inbred family indicates that the trait is recessive. (c) A family in which two albino parents have nonalbino children demonstrates that homozygosity for a recessive allele of either of two genes can cause OCA.

a: © Radu Sighet/Reuters

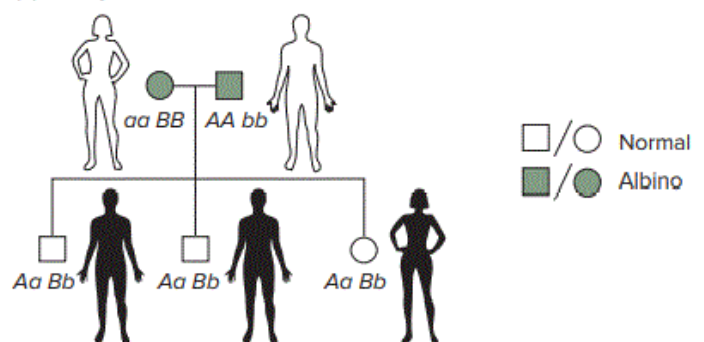
(a) Ocular-cutaneous albinism (OCA)



(b) OCA is recessive



(c) Complementation for albinism



Concept 3.3: Extensions to Mendel for Multifactorial Inheritance

The inheritance of many traits appears to be more complex than can be explained by the participation of only one or two genes in patterns compatible with straightforward Mendelian principles..why?

- 1- More than two genes can influence certain traits
- 2- genes are not the only players: **environment** and **chance** events can sometimes exert considerable effects on traits that are otherwise genetically determined

➔ **Multifactorial traits: traits determined by several different genes, or by the interaction of genes with the environment.**

The Same Genotype Does Not Always Produce the Same Phenotype

In our discussion of gene interactions so far, we have considered examples in which a genotype reliably fashions a particular phenotype. But this is not always what happens...

➔ Sometimes a genotype is not expressed at all: meaning that even though the genotype is present, the expected phenotype does not appear.

➔ Other times, the trait caused by a genotype is expressed to varying degrees or in a variety of ways in different individuals

- What are the factors that alter the phenotypic expression of the genotype?
 - a. Modifier genes
 - b. Environment
 - c. Chance

Penetrance and expressivity

➔ Penetrance: is the percentage of individuals with a particular genotype that show the expected phenotype

➔ Can be **Complete (100%)** or

Incomplete 0%

هاد المصطلح جدا مهم ورح يضل معكم لبعدين. ال Penetrance تعني وجود او عدم وجود ال phenotype الناتج عن genotype معين..يعني يا عندو يا اما ما عندو وما في حل وسط..ف لما نحكي انو هاد الشخص penetrant لصفة معينة هاد يعني انو عندو ياها وظهر عليه ال phenotype تبعها ولما نحكي انو not penetrant او incompletely penetrant (has reduced penetrance) الصفة يعني ما عندو ياها ابدًا مع انو يكون عندو ال genotype ولكن ما بظهر ال phenotype تبعها

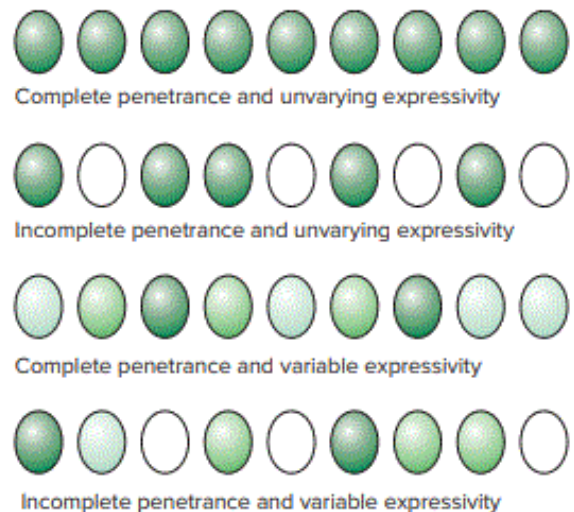
➔ Expressivity: Expressivity is the degree or intensity with which a particular genotype is expressed in a phenotype

➔ Can be **Variable** or **Unvarying**

أما ال expressivity بتشير لمقدار او درجة ال expression ل genotype معين..يعني الها درجات عكس ال penetrance

➔ See Figure 3.25

Figure 3.25 Phenotypes may show variations in penetrance and expressivity. A genotype is completely penetrant when all individuals with that genotype have the same phenotype (green). Some genotypes are incompletely penetrant—some individuals with the same genotype show the phenotype and others do not. Genotypes may also show variable expressivity, meaning that individuals with the same genotype may show the trait but to different degrees.



- Example: Retinoblastoma, the most malignant form of eye cancer, arises from a dominant mutation in one gene, but only about 75% of people who carry the mutant allele develop the disease→ This means that it has a penetrance of 75%

In some people with retinoblastoma, only one eye is affected, while in other individuals with the phenotype, both eyes are diseased→ this means that it has a variable expressivity

- ❖ the incomplete penetrance and variable expressivity of retinoblastoma are mainly the result of **chance**, but in other cases, it is **other genes and/or the environment** that cause variations in phenotype

a. Modifier genes: Alter the phenotypes produced by the alleles of other genes.

- Major genes have a large influence, while modifier genes have a more subtle secondary effect
- Scientists sometimes call the set of unknown modifier genes that influence the action of known genes the **Genetic Background**.
- Example: T locus of mice
 - Mutant T allele causes abnormally short tail., but not all mice carrying the T mutation have the same length tail
 - In different strains, the tails can be 75% of normal length, 50% normal length or 10% of normal length.
→ Different inbred strains must carry alternative alleles of a gene that modifies the T mutant phenotype because all members of each inbred line grow the same length tail, no matter what the environment (for example, diet, cage temperature, or bedding) so geneticists conclude it is genes and not the environment or chance that determine the length of a mutant mouse's tail.

b. Environmental effects on phenotype

Temperature is a common element of the environment that can affect phenotype

- Examples:

1- Coat color in Siamese cats

- These cats are homozygous for one of the multiple alleles of a gene that encodes an enzyme catalyzing the production of the dark pigment melanin.
- The form of the enzyme generated by the variant Siamese allele does not function at the cat's normal core body temperature
→ It becomes active only at the **lower temperatures** found in the cat's extremities, where it promotes the production of melanin, which darkens the animal's ears, nose, paws, and tail
So we say → The enzyme is temperature sensitive. Figure 3.26

2- Fruit fly (*Drosophila melanogaster*)

Temperature can affect **survivability**

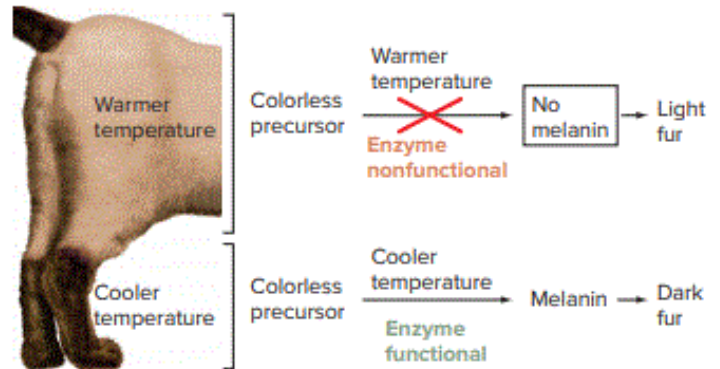
- In one type of experimentally bred fruit fly, some individuals develop and multiply normally at temperatures between 18°C and 29°C
→ but if the thermometer climbs beyond that cutoff for a short time, they become reversibly paralyzed; and if the temperature remains high for more than a few hours, they **die**
- These insects carry a temperature-sensitive allele of the *shibire* gene, which specifies a protein essential for nerve cell transmission.

Figure 3.26 In Siamese cats, temperature affects coat color. (a) A Siamese cat. (b) Melanin is produced only in the cooler extremities. This phenomenon occurs because Siamese cats are homozygous for a mutation that renders an enzyme involved in melanin synthesis temperature sensitive. The mutant enzyme is active at lower temperatures but inactive at higher temperatures.

a: © Renee Lynn/Science Source



(a)



(b)

➔ This type of allele is known as a **Conditional lethal** because it is lethal only under certain conditions.

- Permissive conditions - mutant allele has wild-type functions
- Restrictive conditions - mutant allele has defective functions

➔ Thus, at one temperature, the allele gives rise to a phenotype that is indistinguishable from the wild type, while at another temperature, the same allele generates a mutant phenotype (in this case, lethality). Flies with the wildtype *shibire* allele are viable even at the higher temperatures. The fact that some mutations are lethal only under certain conditions clearly illustrates that the environment can affect the penetrance of a phenotype

- Other effects of environment on phenotype:

Phenocopy: phenotype arising from an environmental agent that mimics the effect of a mutant gene

➔ Not heritable

➔ Can be deleterious or beneficial

Examples in humans:

- Thalidomide produced a phenocopy of phocomelia, a rare dominant trait
- Children with heritable phenylketonuria (PKU) will develop a range of neurological problems, including convulsive seizures and mental impairment, unless they are put on a special diet.
- Genetic predisposition to cardiovascular disease can be influenced by diet and exercise
- Genetic predisposition to lung cancer is strongly affected by cigarette smoking

c. Chance:

The effects of random events on penetrance and expressivity:

Whether a carrier of the retinoblastoma mutation described earlier develops the phenotype, and whether the cancer affects one or both eyes, depend on additional genetic events that occur at random.

To produce retinoblastoma, these events must alter the second allele of the gene in specific body cell

➔ random events that can trigger the onset of the disease include:

- cosmic rays (to which humans are constantly exposed) that alter the genetic material in retinal cells

- or mistakes made during cell division in the retina
- ➔ Chance events provide the **Second hit**—a mutation in the second copy of the retinoblastoma gene—necessary to turn a normal retinal cell into a cancerous one.

The phenotype of retinoblastoma thus results from a specific heritable mutation in a specific gene, but the incomplete penetrance and variable expressivity of the disease depend on random genetic events that affect the other allele in certain cells.

Mendelian Principles Can Also Explain Continuous Variation

Discontinuous traits give clear-cut, "either-or" phenotypic differences between alternative alleles

- Example: All of the traits Mendel studied in peas were discontinuous

Continuous traits are determined by segregating alleles of many genes that interact together and with the environment

- Often appear to blend and "unblend"
- Also called **quantitative traits** because the traits vary over a range that can be measured
- Usually **polygenic** – controlled by multiple genes
- Example: Height and skin color

Mendelian explanation of continuous variation:

- More genes or alleles result in more phenotypic classes and more similarity to continuous variation.
- ➔ In these examples, alleles are **incompletely dominant** and have **additive effects** (Figure 3.28 a)
- More genes or alleles ➔ more phenotypic classes ➔ more similarity to continuous variation. (Figure 3.28 b, c & d)

Figure 3.27 Continuous traits in humans. (a) Women runners at the start of a 5th Avenue mile race in New York City demonstrate that height is a trait showing continuous variation. (b) The skin color of most F_1 offspring is usually between the parental extremes, while the F_2 generation exhibits a broader distribution of continuous variation.

a: © Rudi Von Briel/PhotoEdit

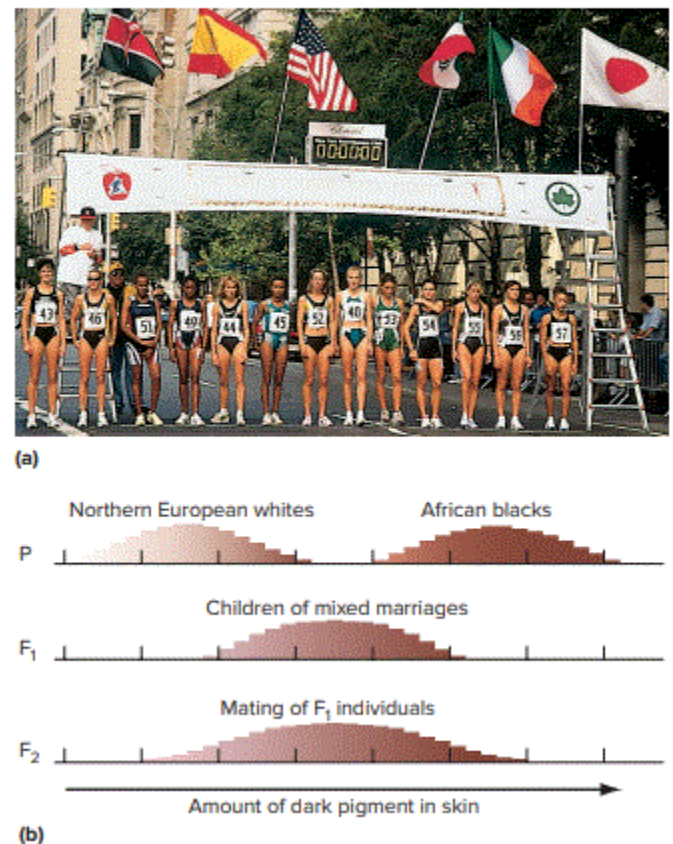


Figure 3.28 A Mendelian explanation of continuous variation. The more genes or alleles, the more possible phenotypic classes, and the greater the similarity to continuous variation. In these examples, several pairs of incompletely dominant alleles have additive effects. Percentages shown at the bottom denote frequencies of each genotype expressed as fractions of the total population.

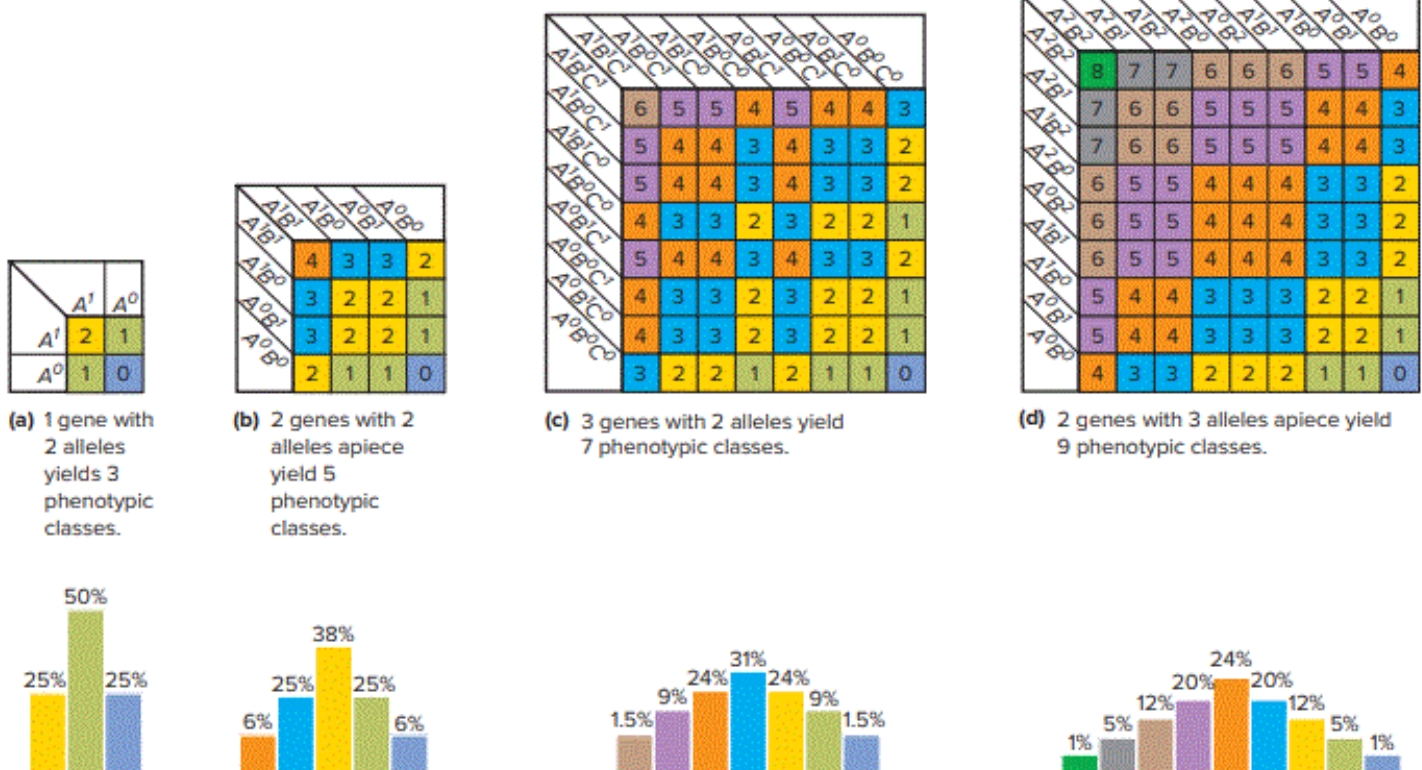


Figure 3.30 Dog color pattern is a polygenic trait. The major alleles determining three kinds of dog coat color patterns. (top): © Tierfotoagentur/Alamy; (middle): © Vanessa Grossemey/Alamy; (bottom): © Martin Rogers/Getty Images



- Example: Dog coat color is determined by multiple alleles of several genes

Genes E and B → Synthesis and deposition of the pigment eumelanin

- See Figure 3.15 (Labrador retrievers)
- A third gene E allele (E^m) results in a dark mask

Genes S and M → Control spotting

- Homozygotes for recessive allele of gene S ($s^p s^p$) are white with colored spots (due to other genes)
- The M gene has two codominant alleles.
 - M^1 phenotype is diluted color
 - M^2 phenotype is normal color

☞ 14 genes control dog coat color and pattern. The roles of seven of these genes—E, B, A, K, D, S, and M—are the best understood.

☞ Some have multiple alleles with incomplete dominance and epistasis

➔ Table 3.3 lists the proteins these genes specify and the nature of various alleles found in domesticated dog

TABLE 3.3 Some of the Genes Affecting Domestic Dog Coat Color and Pattern

Gene	Protein	Dominance Series of Alleles	Phenotypes
Pigment-type switch genes			
Gene <i>A Agouti</i>	Agouti signalling protein (ASIP)	$A^Y > a^w > a^t > a$	A^Y Fawn (lots of light pigment on hair) a^w Agouti (light stripe on dark hair) a^t Tan belly (only hairs on belly have some light pigment) a Black or brown (no light stripe on hairs)
Gene <i>E Extension</i>	Melanocortin receptor (MC1R)	$E^m > E > e$	E^m Black mask on fawn or brindle E Eumelanin (dark) and pheomelanin (yellow) pigments e Only pheomelanin (cream, tan, red)
Gene <i>K Kurokami</i>	Beta-defensin	$K^b > k^{br} > k^y$	K^b Solid color k^{br} Brindled k^y Gene <i>A</i> markings expressed normally
Dilution genes			
Gene <i>D Dilute</i>	Melanophilin (MLPH)	$D > d$	D Colors not dilute d Colors dilute
Gene <i>B Brown</i>	Tyrosine-related protein (TYRP1)	$B > b$	B Black: eumelanin deposited densely b Brown: eumelanin deposited less densely
Pigment cell development and survival genes			
Gene <i>S Spotting</i>	Microphthalmia-associated transcription factor (MITF)	$S > s^p$	S No white markings s^p Colored patches on white background
Gene <i>M Merle</i>	Premelanosome protein (PMEL)	$M^1 = M^2$	M^1 Coat color diluted (homozygote has various health problems) M^2 Normal color